

# Western MA Per Pupil Expenditure Report

With selected districts for comparison

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# Appendix A. Data Sources & Calculation Methodology

## 1. Data sources

### Expenditure dataset

Massachusetts Department of Elementary and Secondary Education (DESE)

District Expenditures by Spending Category

Last updated: August 12, 2025

URL: <https://educationtocareer.data.mass.gov/Finance-and-Budget/District-Expenditures-by-Spending-Category/er3w-dyti/>

This dataset provides:

- Expenditures Per Pupil by category for all Massachusetts school districts
- Student enrollment (In-District FTE, Out-of-District FTE, Total FTE)
- Annual End-of-year-report (EOYR) data from FY2009 to present

### Chapter 70 District Profiles

Massachusetts Department of Elementary and Secondary Education (DESE)

Chapter 70 District Profiles

Retrieved: September 2025

URL: <https://www.doe.mass.edu/finance/chapter70/default.html>

From the Ch70 website: "Chapter 70 District Profiles: The on-line Chapter 70 database shows, for each school district, yearly spending and state aid totals in comparison to the foundation budget. Trend data is available for each year going back to FY1993."

This dataset provides:

- Chapter 70 Aid (c70aid): State aid received by each district
- Required Net School Spending (rqdnss2): Minimum spending required by state law
- Actual Net School Spending (actualNSS): Total district spending on education
- Foundation Enrollment (distfoundenro): District foundation enrollment used for Ch70 funding calculations
- Data reported annually each Fall from FY1993 to present

### District classifications and geography

MassGIS Data: Public School Districts

Polygon data layers depicting the boundaries of public school districts in the Commonwealth of Massachusetts.

School type classifications (Traditional, Regional, Charter, etc.)

Last updated: June 2019

URL: <https://www.mass.gov/info-details/massgis-data-public-school-districts>

MassGIS Data: MA Executive Office of Health & Human Services Regions

Districts classified by region based on EOHHS regional designations.

Last updated: April 2022

URL: <https://www.mass.gov/info-details/massgis-data-ma-executive-office-of-health-human-services-regions>

## 2. Key definitions & calculations

### Per-pupil expenditure (PPE)

PPE is reported by the DESE EOYR. It is calculated with In-District FTE enrollment in the denominator. Per DESE's Researcher's Guide: "Out-of-district costs cannot be properly reported as per-pupil expenditure because tuition costs vary greatly depending on the reason for going out of district."

PPE includes 10 subcategories in the EOYR:

- Administration
- Guidance, Counseling and Testing
- Instructional Leadership

- Instructional Materials, Equipment and Technology
- Insurance, Retirement Programs and Other
- Operations and Maintenance
- Other Teaching Services
- Professional Development
- Pupil Services
- Teachers

This report groups two subcategories in 'Other':

- Instructional Materials, Equipment and Technology
- Professional Development

## Chapter 70 and Net school spending (NSS)

To understand funding sources, we decompose total spending into three stacked components:

- Chapter 70 Aid (\$/pupil):  $\text{State funding} = \text{c70aid} / \text{foundation enrollment}$
- Required local contribution (\$/pupil) =  $\max(0, \text{rqdnss2} - \text{c70aid}) / \text{foundation enrollment}$
- Actual local contribution above required (\$/pupil) =  $(\text{actualNSS} - \text{rqdnss2}) / \text{foundation enrollment}$

Total NSS per foundation pupil = Sum of all three components =  $\text{actualNSS} / \text{foundation enrollment}$

## Note about enrollment denominators

Chapter 70 aid and net school spending calculations use foundation enrollment for per-pupil figures, while PPE totals and expense categories use in-district FTE enrollment. These metrics come from different DESE reporting touchpoints:

- Foundation enrollment (Chapter 70 profile): All students residing in the district on October 1st of the prior year for whom the district is financially responsible.
- In-district FTE (End-of-year-report (EOYR)): Full-year enrollment calculated from total membership days for students either residing in the district, or choiced or tuitioned into the district.

This difference complicates direct comparison between Chapter 70 and expense analyses, but remains valid since each district uses the same denominator type within each metric. The comparison is not apples-to-oranges, but a picture of apples in the morning compared with a picture of apples in the evening. For more information, see the [DESE Researcher's Guide](https://www.doe.mass.edu/research/researchers.html): <https://www.doe.mass.edu/research/researchers.html>

## Compound Annual Growth Rate (CAGR)

CAGR measures the smoothed annual growth rate over multiple years — answering "what steady annual rate would produce the same start-to-end result?"

Formula:  $\text{CAGR} = (\text{End\_Value} / \text{Start\_Value})^{(1 / \text{Years})} - 1$

Example: If expenditures grow from \$10,000 to \$15,000 over 10 years:

$\text{CAGR} = (\$15,000 / \$10,000)^{(1/10)} - 1 = 4.14\% \text{ per year}$

## Weighted average

For cohort aggregates, we use enrollment-weighted averages to ensure larger districts appropriately influence the cohort average:

Formula:  $\text{Weighted\_PPE} = \Sigma(\text{District\_PPE} \times \text{District\_FTE}) / \Sigma(\text{District\_FTE})$

## 3. Enrollment-based cohort methodology

Districts are grouped by FY2024 In-District FTE enrollment using quartile analysis:

### Cohort boundaries (FY2024)

Based on 59 Western MA traditional districts:

- Q1 (25th percentile): 154.7 FTE → rounded to 200
- Median (50th percentile): 792.0 FTE → rounded to 800
- Q3 (75th percentile): 1,597.6 FTE → rounded to 1,600
- Outlier threshold: Fixed at 10,000 FTE

Final cohorts:

- **Tiny (0-200 FTE):** 17 districts (Cohort 1: below Q1) **Member districts:** conway, erving, farmington river reg, florida, hancock, hawlemont, leverett, new salem-wendell, pelham, petersham, richmond, rowe, savoy, shutesbury, sunderland, and 2 others
- **Small (201-800 FTE):** 13 districts (Cohort 2: Q1 to median) **Member districts:** clarksburg, deerfield, frontier, gateway, granby, hadley, hatfield, lee, lenox, orange, pioneer valley, ralph c mahar, southern berkshire
- **Medium (801-1600 FTE):** 15 districts (Cohort 3: median to Q3) **Member districts:** amherst, amherst-pelham, berkshire hills, central berkshire, easthampton, gill-montague, greenfield, hoosac valley regional, mohawk trail, monson, mount greylock, north adams, palmer, southwick-tolland-granville regional school district, ware
- **Large (1601-10K FTE):** 14 districts (Cohort 4: above Q3) **Member districts:** agawam, athol-royalston, belchertown, chicopee, east longmeadow, hampden-wilbraham, holyoke, longmeadow, ludlow, northampton, pittsfield, south hadley, west springfield, westfield
- **Outliers (Springfield >10K FTE):** 1 district (Cohort 5: statistical outlier, analyzed separately) **Member districts:** springfield

## 4. Comparison shading methodology

To visualize meaningful differences while filtering statistical noise, we use graduated color shading based on two independent tests:

### Colors

- Orange: District exceeds baseline
- Blue: District below baseline
- White: Within threshold (not meaningfully different)

### Test 1: Dollar amounts (2024 values)

- Threshold:  $\geq 5.0\%$  difference from baseline
- Calculation:  $(\text{District} - \text{Baseline}) / \text{Baseline}$
- Shading intensity bins: 5%, 10%, 15%, 20%+

### Test 2: Growth rates (CAGR)

- Threshold:  $\geq 0.5$  percentage points (pp) difference from baseline
- Calculation:  $\text{District\_CAGR} - \text{Baseline\_CAGR}$
- Shading intensity bins: 0pp, 1pp, 2pp, 2pp+

### Statistical justification for thresholds

Analysis of all 59 Western MA districts revealed different natural variation by metric:

- PPE variation: Mean = \$24,237, SD = \$5,462, CV = 22.5%
- CAGR variation: Mean = 6.00pp, SD = 3.24pp, CV = 54.0%

The 5.0%/0.5pp thresholds balance statistical rigor with practical communication:

- 5.0% for PPE: Flags differences  $\geq \$1,212$  at mean ( $\sim 0.22$  SD)
- 0.5pp for CAGR: Flags differences  $\geq 0.5$ pp ( $\sim 0.15$  SD)
- Balance ratio:  $0.7\times$  (appropriately sensitive to compound growth effects)

This approach creates a visual hierarchy:

- No shading: Districts statistically similar to peers
- Lighter shading: Notable difference
- Darker shading: Exceptional difference requiring attention

These thresholds reflect the different ways these metrics behave. PPE and enrollment are dollar amounts and headcounts — a

5.0% difference is substantial and visible in annual budgets. But CAGR measures rates of change, and small differences in growth rates compound dramatically over time. Consider a district with \$20,000 PPE:

- Growing at 4.0% annually reaches \$36,019 after 15 years
- Growing at 4.5% annually reaches \$38,318 after 15 years
- That 0.5pp difference produces a \$2,299 gap (11.5% more total growth)

For a 300-student district, that's \$689,700 in additional annual spending by year 15. For municipalities managing long-term budgets and multi-year labor contracts, these trajectories matter—even when year-to-year changes seem modest.

The tighter CAGR threshold (0.5pp vs. 5%) also reflects statistical reality: growth rates vary more widely across districts (CV = 54%) than PPE levels do (CV = 22.5%). A 0.5pp threshold flags districts with growth trajectories meaningfully different from their cohort without overcounting normal variation. By using a more sensitive threshold for CAGR, the analysis captures the real-world policy significance of compound growth differences while maintaining appropriate sensitivity for dollar and enrollment comparisons.

## 5. Calculation Example: Baseline vs. Growth Analysis

To illustrate how historical baselines drive current spending differences, consider this detailed example comparing a district to its cohort average:

### Amherst-Pelham Regional (Medium Cohort)

Starting point (2009):

- District PPE: \$16,211 per pupil
- Medium cohort average: \$13,032 per pupil
- Dollar difference:  $\$16,211 - \$13,032 = \$3,179$  above cohort
- Percentage difference:  $(\$3,179 \div \$13,032) \times 100 = 24.4\%$  above cohort

Calculating 15-year growth rates (2009-2024):

District CAGR:

- Formula:  $\text{CAGR} = (\text{End\_Value} \div \text{Start\_Value})^{(1 \div \text{Years})} - 1$
- $\text{CAGR} = (\$28,233 \div \$16,211)^{(1/15)} - 1 = (1.74139)^{(0.06667)} - 1 = 1.0377 - 1 = 3.77\%$  annually

Medium cohort CAGR:

- $\text{CAGR} = (\$23,785 \div \$13,032)^{(1/15)} - 1 = (1.82525)^{(0.06667)} - 1 = 1.0409 - 1 = 4.09\%$  annually

Growth rate difference:  $3.77\% - 4.09\% = -0.32$  percentage points (District grew slower than cohort)

Ending point (2024):

- District PPE: \$28,233 per pupil
- Medium cohort average: \$23,785 per pupil
- Dollar difference:  $\$28,233 - \$23,785 = \$4,448$  above cohort
- Percentage difference:  $(\$4,448 \div \$23,785) \times 100 = 18.7\%$  above cohort

### Key insight

Despite growing more slowly than peer districts (3.77% vs. 4.09% annually), Amherst-Pelham remains well above the cohort average because it started from a higher baseline. The district actually narrowed the gap from 24.4% to 18.7% above peers — evidence of relative fiscal restraint over 15 years, not excess growth. The absolute dollar difference grew (\$3,179 → \$4,448) because both the district and cohort experienced inflation and cost increases, but the relative position improved, showing the district's growth has been modest compared to peers.

This pattern — where high current spending reflects high historical baselines rather than high growth — appears across multiple Western MA districts.

## 6. Weighted Average Calculation Example: Medium Cohort

This example shows the complete weighted average calculation for the Medium cohort's 2024 Teachers category PPE using all 15 member districts.

## Medium Cohort Member Districts (2024 In-District FTE)

1. Mohawk Trail: 804.8 FTE
2. Monson: 807.8 FTE
3. Gill-Montague: 899.8 FTE
4. Amherst: 1,001.5 FTE (Teachers PPE: \$9,745 from Appendix B. Table 22)
5. Hoosac Valley Regional: 1,002.7 FTE
6. Palmer: 1,061.5 FTE
7. Ware: 1,095.8 FTE
8. Berkshire Hills: 1,174.4 FTE
9. North Adams: 1,196.0 FTE
10. Mount Greylock: 1,206.9 FTE
11. Amherst-Pelham: 1,209.3 FTE (Teachers PPE: \$8,270 from Appendix B. Table 19)
12. Southwick-Tolland-Granville Regional: 1,312.8 FTE
13. Greenfield: 1,394.0 FTE
14. Easthampton: 1,403.7 FTE
15. Central Berkshire: 1,585.9 FTE

Total Medium Cohort Enrollment: 17,156.9 FTE

## Weighted Average Calculation for Teachers PPE (2024)

For each district, we multiply their Teachers PPE by their enrollment weight:

[District PPE × (District FTE ÷ 17,156.9)]

The sum of all 15 weighted contributions equals the Medium cohort weighted average: \$7,431 per pupil for Teachers expenditure in 2024.

## Verifying with Total PPE (2024)

Medium Cohort Total PPE components sum to:

- Other: \$1,200
- Guidance, Counseling and Testing: \$826
- Administration: \$911
- Instructional Leadership: \$1,573
- Operations and Maintenance: \$1,953
- Other Teaching Services: \$2,395
- Pupil Services: \$2,755
- Insurance, Retirement and Other: \$4,740
- Teachers: \$7,431
- Total: \$23,785 per pupil

## Cohort CAGR Calculation (2009-2024)

Using the weighted totals for all 15 districts at the starting point (2009):

- Total enrollment: 21,996 FTE (15 districts × avg 1,466 from Table 8)
- Weighted Total PPE: \$13,032 per pupil

Ending point (2024):

- Total enrollment: 17,156.9 FTE (sum of all 15 districts)
- Weighted Total PPE: \$23,785 per pupil

CAGR calculation:

- $CAGR = (\$23,785 \div \$13,032)^{(1/15)} - 1 = CAGR = (1.82525)^{(0.06667)} - 1 = 1.0409 - 1 = 4.09\%$  annually

## Key insights:

- Enrollment decline: The Medium cohort's total enrollment dropped from ~22,000 to ~17,000 students (-22%) over 15 years
- Weighting impact: Central Berkshire (1,586 FTE) has nearly twice the influence of Mohawk Trail (805 FTE) in the weighted average

## Appendix B. Statistical Associations

### Key findings

Statistical tests reveal that commonly cited factors — enrollment size, enrollment change, PPE growth rates, and Chapter 70 aid patterns — show weak associations with current per-pupil expenditure levels, explaining less than 15% of spending variation. In contrast, two factors dominate: historical baseline PPE (what districts were spending in 2009) and local contribution above required (Actual NSS above Required NSS), which together explain over 70% of spending differences. Districts maintain their relative spending positions over time regardless of enrollment trends, state aid changes, or growth rates. This persistent pattern of dependency on legacy spending and local contribution suggests that effective school budgeting requires more attention to district-specific histories and community decisions, and less attention to broad generalizations about structural factors.

### Statistical tests used in this analysis

This appendix presents exploratory statistical tests examining associations between budget and enrollment variables and per-pupil expenditure levels in 59 Western Massachusetts traditional school districts. The analyses below are not meant to imply that this report has conducted rigorous econometric research. Rather, these statistical tests are practical and replicable ways of identifying stronger or weaker relationships between school spending and variables in this report. However, other variables not considered in this report are also worthy of close inspection. All correlations and associations presented should be interpreted as descriptive rather than predictive or prescriptive.

Throughout this analysis, we use  $p < 0.05$  as the threshold for statistical significance.

Here's a brief summary of each statistical test and its purpose:

#### Linear Regression ( $R^2$ and coefficients)

Shows how much of the variation in PPE can be explained by a single factor. An  $R^2$  of 0.74 means that factor explains 74% of spending differences, while an  $R^2$  of 0.03 means it explains almost nothing. This helps identify which factors actually matter versus those that just seem important.

#### Correlation Analysis by Cohort

Tests whether relationships hold within groups of similarly-sized districts. If enrollment factors "cause" spending patterns, we'd see consistent correlations within each size cohort. When correlations disappear within cohorts, it suggests the overall pattern may be spurious or coincidental.

#### Analysis of Variance (ANOVA)

Tests whether districts grouped by a characteristic (like high vs. low state aid) have significantly different average PPE. Helps determine if categories that seem meaningful actually create distinct spending patterns. The between-group vs. within-group variance ratio shows whether differences between categories exceed normal variation within them.

#### Post-hoc Tukey HSD Test

When ANOVA finds differences, this identifies which specific groups differ from each other. Prevents over-interpreting by controlling for multiple comparisons — important when checking many possible differences simultaneously.

#### Multiple Regression Models

Tests how factors work together to explain the dependent variable — PPE, in this case. Reveals which variables remain important when controlling for others, and which lose their apparent influence. Helps distinguish genuine predictors from those that merely correlate with the real drivers.

### 1. Enrollment and PPE

This section provides statistical evidence supporting the report's finding that district enrollment size has limited explanatory power for per-pupil expenditure variations.

#### Linear Regression Analysis

- Model:  $PPE(2024) = \beta_0 + \beta_1 \times \ln(\text{Enrollment}) + \varepsilon$
- Using natural log of enrollment to account for non-linear relationship between size and costs.

## Results for 59 Western MA traditional districts

- $R^2 = 0.22$
- Adjusted  $R^2 = 0.21$
- F-statistic: 16.2 ( $p < 0.001$ )
- Standard error: \$4,892

## Interpretation

Enrollment size explains only 22% of the variation in per-pupil expenditures. The remaining 78% is attributable to other factors not captured by enrollment alone.

## Correlation analysis by cohort

Pearson correlation coefficients between enrollment and PPE within each cohort:

### Tiny (0-200): No significant relationship.

- $n = 17$
- Correlation = -0.18
- p-value = 0.49.

### Small (201-800): No significant relationship.

- $n = 13$
- Correlation = 0.23
- p-value = 0.45

### Medium (801-1600): No significant relationship.

- $n = 15$
- Correlation = 0.31
- p-value = 0.26

### Large (1601-10K): No significant relationship.

- $n = 14$
- Correlation = 0.42
- p-value = 0.13

## Key finding

Within enrollment cohorts, there is no statistically significant correlation between size and spending. This suggests that factors other than enrollment drive spending differences even among similarly-sized districts.

## Analysis of Variance (ANOVA)

Testing whether mean PPE differs significantly across enrollment cohorts.

### One-way ANOVA results

- $F(4, 54) = 7.82, p < 0.001$
- Between-group variance: 22%
- Within-group variance: 78%

### Post-hoc Tukey HSD test:

- Tiny vs Large: Mean difference = \$6,663 ( $p < 0.01$ )
- Tiny vs Medium: Mean difference = \$3,719 ( $p = 0.08$ )



- Small vs Large: Mean difference = \$3,781 ( $p = 0.04$ )
- Other pairwise comparisons: not significant ( $p > 0.10$ )

### Interpretation

While cohorts show statistically different means, the within-cohort variation (78%) far exceeds between-cohort variation (22%), confirming that enrollment cohort is a weak predictor of spending.

## Growth rate analysis

### Correlation between enrollment change and PPE growth (2009-2024)

- Correlation coefficient:  $r = -0.14$
- p-value: 0.29

### Interpretation

No significant relationship between enrollment trends and spending growth. This finding contradicts the common assumption that declining enrollment drives cost increases. Districts with stable, growing, and declining enrollment all show similar PPE growth rates.

## Multiple regression model

Extended Model:  $\text{PPE (2024)} = \beta_0 + \beta_1 \times \ln(\text{Enrollment}) + \beta_2 \times \text{PPE(2009)} + \beta_3 \times \text{ActualNSS} + \varepsilon$   
Where ActualNSS = Actual NSS above Required NSS per pupil

### Results

- $R^2 = 0.71$
- Adjusted  $R^2 = 0.69$
- Variables' contribution to  $R^2$ :
  - Historical PPE (2009): 49%
  - Local funding effort: 20%
  - Enrollment: 2%

### Interpretation

When controlling for historical spending patterns and local funding decisions, enrollment becomes nearly irrelevant to explaining current PPE levels. Historical baselines (where districts started) and local choices (how much communities invest above requirements) are the primary drivers.

## Conclusion

District enrollment size, while correlated with spending at the aggregate level, explains relatively little about why specific districts spend what they do. The large within-cohort variations relative to between-cohort differences consistently indicate that enrollment-based categorization, while administratively useful, has limited predictive power for expenditure levels.

## 2. Statistical Analysis of PPE Growth-Current PPE Relationships

This section examines whether districts with faster spending growth over 15 years have higher current per-pupil expenditures.

### Linear regression analysis

- Model:  $\text{PPE (2024)} = \beta_0 + \beta_1 \times \text{PPE CAGR (2009-2024)} + \varepsilon$
- Testing whether growth rates predict current spending levels.

### Results for 59 Western MA traditional districts

- $R^2 = 0.09$
- Adjusted  $R^2 = 0.07$
- F-statistic: 5.4 ( $p = 0.024$ )
- Standard error: \$5,309

### Interpretation

PPE growth rate explains only 9% of the variation in current per-pupil expenditures. The remaining 91% reflects that starting points matter far more than growth rates in determining current spending levels.

### Correlation analysis by cohort

Pearson correlation coefficients between PPE CAGR and PPE (2024) within each cohort:

#### Tiny (0-200): No significant relationship.

- $n = 17$
- Correlation = 0.21
- p-value = 0.42

#### Small (201-800): No significant relationship.

- $n = 13$
- Correlation = -0.14
- p-value = 0.65

#### Medium (801-1600): No significant relationship.

- $n = 15$
- Correlation = 0.08
- p-value = 0.77

#### Large (1601-10K): No significant relationship.

- $n = 14$
- Correlation = 0.19
- p-value = 0.51

### Key finding

Within enrollment cohorts, there is no correlation between growth rates and current spending levels. High-spending districts include both fast and slow growers, confirming that historical baselines dominate current positions.

### Analysis of Variance (ANOVA)

Testing whether mean PPE differs across growth rate quartiles.

#### One-way ANOVA results

- $F(3, 55) = 2.31, p = 0.09$
- Between-group variance: 11%
- Within-group variance: 89%

#### Post-hoc Tukey HSD test:

- All pairwise comparisons: not significant ( $p > 0.10$ )

### Interpretation

Growth rate quartiles show no statistically significant differences in mean PPE. The overwhelming within-group variation (89%) confirms that these 15-year growth patterns don't determine current spending positions.

## Historical baseline analysis

### Correlation between PPE (2009) and PPE (2024)

- Correlation coefficient:  $r = 0.87$
- p-value:  $< 0.001$

### Interpretation

The extremely strong correlation between 2009 and 2024 spending levels demonstrates path dependency. Where districts started matters far more than how fast they grew.

### Multiple regression model

Extended Model:  $\text{PPE (2024)} = \beta_0 + \beta_1 \times \text{PPE CAGR} + \beta_2 \times \text{PPE(2009)} + \beta_3 \times \ln(\text{Enrollment}) + \varepsilon$

### Results

- $R^2 = 0.83$
- Adjusted  $R^2 = 0.82$
- Variables' contribution to  $R^2$ :
  - Historical PPE (2009): 76%
  - PPE growth rate: 5%
  - Current enrollment: 2%

### Interpretation

When controlling for starting position, growth rates add minimal explanatory power. Districts maintain their relative spending positions over time despite varying growth rates.

### Note about statistical significance

The weak relationship between growth rates and current levels highlights that percentage growth can be misleading — a 3% annual increase from \$15,000 adds far more dollars than 5% growth from \$10,000.

### Conclusion

Current PPE levels are overwhelmingly determined by historical baselines rather than recent growth rates. High-spending districts today were generally high-spending districts 15 years ago, regardless of growth patterns.

## 3. Statistical Analysis of Chapter 70 Aid-PPE Relationships

This section examines the relationship between state Chapter 70 aid levels and total per-pupil expenditures.

### Linear Regression Analysis

- Model:  $\text{PPE (2024)} = \beta_0 + \beta_1 \times \text{Chapter 70 per pupil (2024)} + \varepsilon$
- Testing whether higher state aid correlates with higher total spending.

### Results for 59 Western MA traditional districts

- $R^2 = 0.03$
- Adjusted  $R^2 = 0.01$
- F-statistic: 1.7 ( $p = 0.20$ )
- Standard error: \$5,479

### Interpretation

Chapter 70 aid levels explain only 3% of the variation in total per-pupil expenditures. State aid and total spending are essentially uncorrelated, suggesting that local funding decisions offset state aid patterns.

## **Correlation analysis by cohort**

Pearson correlation coefficients between Chapter 70 per pupil and PPE within each cohort:

### **Tiny (0-200): No significant relationship.**

- $n = 17$
- Correlation = -0.09
- p-value = 0.73

### **Small (201-800): No significant relationship.**

- $n = 13$
- Correlation = 0.11
- p-value = 0.72

### **Medium (801-1600): No significant relationship.**

- $n = 15$
- Correlation = -0.16
- p-value = 0.57

### **Large (1601-10K): No significant relationship.**

- $n = 14$
- Correlation = 0.08
- p-value = 0.78

## **Key finding**

Within enrollment cohorts, Chapter 70 aid shows no correlation with total spending. Districts receiving more state aid do not predictably spend more or less per pupil than their cohort average.

## **Analysis of Variance (ANOVA)**

Testing whether mean PPE differs across Chapter 70 aid quartiles.

### **One-way ANOVA results**

- $F(3, 55) = 1.89$ ,  $p = 0.14$
- Between-group variance: 9%
- Within-group variance: 91%

### **Post-hoc Tukey HSD test:**

- All pairwise comparisons: not significant ( $p > 0.10$ )

## **Interpretation**

Chapter 70 aid quartiles show no statistically significant differences in mean total PPE. High-aid districts span the full range of total spending levels.

## **Offset pattern analysis**

### **Correlation between Chapter 70 and local contribution above requirement**

- Correlation coefficient:  $r = -0.52$
- p-value:  $< 0.001$

## Interpretation

The significant negative correlation reveals an offset pattern: communities receiving more state aid tend to invest less above their required minimums, partially neutralizing the equalizing intent of state aid.

## Multiple regression model

Extended Model:  $PPE(2024) = \beta_0 + \beta_1 \times Ch70PerPupil + \beta_2 \times LocalInvestment + \beta_3 \times \ln(Enrollment) + \varepsilon$

## Results

- $R^2 = 0.68$
- Adjusted  $R^2 = 0.66$
- Variables' contribution to  $R^2$ :
  - Local contribution: 56%
  - Chapter 70 aid: 9%
  - Current enrollment: 3%

## Interpretation

Local contribution decisions dominate total spending determination. When controlling for local contribution, Chapter 70 aid has modest positive effect, but the offsetting behavior limits its impact on total PPE.

## Note about statistical significance

The lack of correlation between state aid and total spending suggests that the Chapter 70 formula's equalizing goals are undermined by local contribution responses that maintain spending disparities.

## Conclusion

Chapter 70 aid levels have virtually no relationship with total PPE due to offsetting local funding decisions. Communities adjust their local contributions based on state aid received, maintaining historical spending patterns regardless of aid levels.

## 4. Statistical Analysis of Chapter 70 Growth-PPE Relationships

This section examines whether districts with faster Chapter 70 aid growth have higher current per-pupil expenditures.

### Linear Regression Analysis

- Model:  $PPE(2024) = \beta_0 + \beta_1 \times \text{Chapter 70 CAGR (2009-2024)} + \varepsilon$
- Testing whether state aid growth rates predict current total spending.

### Results for 59 Western MA traditional districts

- $R^2 = 0.01$
- Adjusted  $R^2 = -0.01$
- F-statistic: 0.6 ( $p = 0.44$ )
- Standard error: \$5,531

## Interpretation

Chapter 70 growth rate explains less than 1% of the variation in current per-pupil expenditures. State aid growth patterns have no meaningful relationship with total spending levels.

## Correlation analysis by cohort

Pearson correlation coefficients between Chapter 70 CAGR and PPE within each cohort:

**Tiny (0-200): No significant relationship.**

- n = 17
- Correlation = 0.05
- p-value = 0.85

#### **Small (201-800): No significant relationship.**

- n = 13
- Correlation = -0.08
- p-value = 0.79

#### **Medium (801-1600): No significant relationship.**

- n = 15
- Correlation = 0.12
- p-value = 0.67

#### **Large (1601-10K): No significant relationship.**

- n = 14
- Correlation = -0.03
- p-value = 0.92

#### **Key finding**

Within all enrollment cohorts, Chapter 70 growth shows zero correlation with current spending. Districts experiencing rapid aid growth are no more likely to have high PPE than those with stagnant aid.

#### **Analysis of Variance (ANOVA)**

Testing whether mean PPE differs across Chapter 70 growth rate quartiles.

##### **One-way ANOVA results**

- $F(3, 55) = 0.73$ ,  $p = 0.54$
- Between-group variance: 4%
- Within-group variance: 96%

##### **Post-hoc Tukey HSD test:**

- All pairwise comparisons: not significant ( $p > 0.10$ )

#### **Interpretation**

Chapter 70 growth quartiles show identical mean PPE levels. The minimal between-group variance (4%) confirms that state aid growth is irrelevant to current spending positions.

#### **Local offset analysis**

##### **Correlation between Chapter 70 CAGR and local contribution CAGR**

- Correlation coefficient:  $r = -0.31$
- p-value: 0.02

#### **Interpretation**

The negative correlation suggests communities partially offset state aid increases with slower local contribution growth, dampening the impact on total spending.

#### **Multiple regression model**

Extended Model:  $PPE(2024) = \beta_0 + \beta_1 \times Ch70CAGR + \beta_2 \times LocalInvestCAGR + \beta_3 \times PPE(2009) + \epsilon$

## Results

- $R^2 = 0.79$
- Adjusted  $R^2 = 0.78$
- Variables' contribution to  $R^2$ :
  - Historical PPE (2009): 69%
  - Local contribution growth: 9%
  - Chapter 70 growth: 1%

## Interpretation

Historical spending levels dominate current positions. Local contribution growth has modest impact. Chapter 70 growth is essentially irrelevant to explaining current PPE levels.

## Conclusion

Chapter 70 aid growth rates have no predictive power for current PPE levels. State efforts to increase aid have not altered the fundamental spending patterns across districts, which remain anchored to historical baselines and local decisions. The 2019 Student Opportunity Act's impact may have been dampened if communities adjusted local funding to maintain relative positions.

## 5. Statistical Analysis of Local Investment-PPE Relationships

This section examines the relationship between local contribution (Actual NSS above Required NSS) and total per-pupil expenditures.

### Linear Regression Analysis

- Model:  $PPE(2024) = \beta_0 + \beta_1 \times \text{Local Investment per pupil}(2024) + \epsilon$
- Testing whether higher local effort translates directly to higher total spending.

### Results for 59 Western MA traditional districts

- $R^2 = 0.74$
- Adjusted  $R^2 = 0.73$
- F-statistic: 159.8 ( $p < 0.001$ )
- Standard error: \$2,853

### Interpretation

Local contribution explains 74% of the variation in total per-pupil expenditures. Only historical PPE baselines show stronger predictive power, confirming that current spending reflects the combination of where districts started and how much communities contribute above required NSS.

### Correlation analysis by cohort

Pearson correlation coefficients between local contribution and PPE within each cohort:

#### Tiny (0-200): Strong positive relationship.

- $n = 17$
- Correlation = 0.88
- p-value < 0.001

#### Small (201-800): Strong positive relationship.

- $n = 13$
- Correlation = 0.91
- p-value < 0.001

#### Medium (801-1600): Strong positive relationship.

- n = 15
- Correlation = 0.85
- p-value < 0.001

**Large (1601-10K): Strong positive relationship.**

- n = 14
- Correlation = 0.82
- p-value < 0.001

### Key finding

Within every enrollment cohort, local contribution shows strong positive correlation with total spending. This consistent relationship across all district sizes confirms the level of actual NSS above required NSS as a primary predictor of PPE.

## Analysis of Variance (ANOVA)

Testing whether mean PPE differs across local contribution quartiles.

### One-way ANOVA results

- $F(3, 55) = 41.2$ ,  $p < 0.001$
- Between-group variance: 69%
- Within-group variance: 31%

### Post-hoc Tukey HSD test:

- Q4 vs Q1: Mean difference = \$9,847 ( $p < 0.001$ )
- Q4 vs Q2: Mean difference = \$7,234 ( $p < 0.001$ )
- Q3 vs Q1: Mean difference = \$5,621 ( $p < 0.001$ )
- All comparisons: significant ( $p < 0.05$ )

### Interpretation

Local contribution quartiles show highly significant differences in PPE. The strong between-group variance (69%) confirms that local contribution effectively predicts spending levels.

## Marginal effect analysis

### Regression coefficient interpretation

- Each additional \$1,000 in actual NSS above required NSS per pupil increases PPE by \$1,243
- 95% Confidence interval: [\$1,148, \$1,338]

## Multiple regression model

Extended Model:  $PPE(2024) = \beta_0 + \beta_1 \times LocalInvestment + \beta_2 \times Ch70PerPupil + \beta_3 \times \ln(Enrollment) + \varepsilon$

### Results

- $R^2 = 0.81$
- Adjusted  $R^2 = 0.80$
- Variables' contribution to  $R^2$ :
  - Local contribution: 71%
  - Chapter 70 aid: 8%
  - Current enrollment: 2%

### Interpretation



Actual local contribution dominates the model. When communities invest more above requirements, total spending increases nearly dollar-for-dollar regardless of state aid or enrollment size.

## Conclusion

Local contribution above required minimums is the dominant predictor of total PPE, explaining nearly three-quarters of spending variation. Communities that choose to fund above requirements drive the spending disparities observed across districts.

## 6. Statistical Analysis of Local Investment Growth-PPE Relationships

This section examines whether districts with faster local contribution growth have higher current per-pupil expenditures.

### Linear Regression Analysis

- Model:  $PPE(2024) = \beta_0 + \beta_1 \times \text{Actual NSS above Required NSS CAGR (2009-2024)} + \varepsilon$
- Testing whether growth in local effort predicts current total spending.

### Results for 59 Western MA traditional districts

- $R^2 = 0.14$
- Adjusted  $R^2 = 0.12$
- F-statistic: 9.1 ( $p = 0.004$ )
- Standard error: \$5,166

### Interpretation

Local contribution growth rate explains only 14% of the variation in current per-pupil expenditures. While statistically significant, growth in local contribution is a weak predictor of current spending levels.

### Correlation analysis by cohort

Pearson correlation coefficients between local contribution CAGR and PPE within each cohort:

#### Tiny (0-200): No significant relationship.

- $n = 17$
- Correlation = 0.29
- p-value = 0.26

#### Small (201-800): No significant relationship.

- $n = 13$
- Correlation = 0.18
- p-value = 0.56

#### Medium (801-1600): No significant relationship.

- $n = 15$
- Correlation = 0.24
- p-value = 0.39

#### Large (1601-10K): No significant relationship.

- $n = 14$
- Correlation = 0.31
- p-value = 0.28

### Key finding

Within enrollment cohorts, local contribution growth shows no significant correlation with current spending. Districts with rapid local funding increases don't necessarily end up as high spenders if they started from low baselines.

## Analysis of Variance (ANOVA)

Testing whether mean PPE differs across local contribution growth quartiles.

### One-way ANOVA results

- $F(3, 55) = 3.47$ ,  $p = 0.022$
- Between-group variance: 16%
- Within-group variance: 84%

### Post-hoc Tukey HSD test:

- Q4 vs Q1: Mean difference = \$3,892 ( $p = 0.04$ )
- Other pairwise comparisons: not significant ( $p > 0.10$ )

### Interpretation

While the fastest-growing quartile shows modestly higher PPE, the overwhelming within-group variation (84%) confirms that growth rates poorly predict current positions.

## Starting point analysis

### Correlation between local contribution (2009) and local contribution CAGR

- Correlation coefficient:  $r = -0.43$
- p-value:  $< 0.001$

### Interpretation

The negative correlation reveals convergence: districts with lower initial local contribution grew faster, partially closing gaps, but not enough to overcome historical differences.

## Multiple regression model

Extended Model:  $PPE(2024) = \beta_0 + \beta_1 \times LocalInvestCAGR + \beta_2 \times LocalInvest(2009) + \beta_3 \times PPE(2009) + \epsilon$

### Results

- $R^2 = 0.84$
- Adjusted  $R^2 = 0.83$
- Variables' contribution to  $R^2$ :
  - Historical PPE (2009): 62%
  - Historical actual NSS above required (2009): 14%
  - Local contribution growth: 8%

### Interpretation

Historical positions dominate current spending levels. Local contribution growth adds modest explanatory power but cannot overcome the persistence of historical patterns.

## Conclusion

Local contribution growth rates have limited predictive power for current PPE levels. While faster-growing districts show slightly higher spending, historical baselines remain the dominant factor. Communities maintain relative spending positions despite varying growth rates in local effort.

## Key findings

Statistical tests reveal that commonly cited factors — enrollment size, enrollment change, PPE growth rates, and Chapter 70 aid patterns — show weak associations with current per-pupil expenditure levels, explaining less than 15% of spending variation. In contrast, two factors dominate: historical baseline PPE (what districts were spending in 2009) and local contribution above required (Actual NSS above Required NSS), which together explain over 70% of spending differences. Districts maintain their relative spending positions over time regardless of enrollment trends, state aid changes, or growth rates. This persistent pattern of dependency on legacy spending and local contribution suggests that effective school budgeting requires more attention to district-specific histories and community decisions, and less attention to broad generalizations about structural factors.

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## Appendix C. Data Tables

### All data values used in plots

This appendix contains the underlying data tables for all districts and regions shown in the report. Each table shows PPE by category (in \$/in-district FTE pupil), FTE enrollment counts, and NSS/Ch70 funding components (in \$/foundation enrollment pupil) across all available years.

### Data: All Western MA Traditional Districts: Tiny (0-200 FTE)

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

#### PPE by Category (\$/in-district FTE pupil) — Weighted Averages

| Table 1                          |          |          |          |          |          |          |          |          |          |          |          |          |          |          |          |          |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
| Other                            | \$577    | \$571    | \$623    | \$643    | \$647    | \$665    | \$676    | \$709    | \$780    | \$579    | \$673    | \$738    | \$1,380  | \$1,042  | \$1,304  | \$1,057  |
| Guidance, Counseling and Testing | \$345    | \$365    | \$375    | \$402    | \$419    | \$462    | \$475    | \$454    | \$445    | \$448    | \$459    | \$488    | \$577    | \$663    | \$621    | \$674    |
| Administration                   | \$770    | \$808    | \$814    | \$853    | \$915    | \$851    | \$774    | \$819    | \$863    | \$869    | \$770    | \$833    | \$950    | \$1,003  | \$1,114  | \$1,126  |
| Instructional Leadership         | \$1,042  | \$1,096  | \$1,125  | \$1,225  | \$1,269  | \$1,316  | \$1,496  | \$1,584  | \$1,510  | \$1,524  | \$1,543  | \$1,589  | \$1,817  | \$1,733  | \$1,777  | \$1,991  |
| Operations and Maintenance       | \$1,374  | \$1,312  | \$1,460  | \$1,416  | \$1,359  | \$1,558  | \$1,554  | \$1,452  | \$1,433  | \$1,468  | \$1,811  | \$1,670  | \$2,060  | \$2,389  | \$2,232  | \$2,652  |
| Other Teaching Services          | \$1,899  | \$1,926  | \$1,987  | \$2,112  | \$2,412  | \$2,378  | \$2,494  | \$2,635  | \$2,673  | \$2,707  | \$3,004  | \$3,115  | \$3,322  | \$3,677  | \$3,638  | \$3,757  |
| Pupil Services                   | \$1,324  | \$1,427  | \$1,426  | \$1,497  | \$1,605  | \$1,599  | \$1,670  | \$1,753  | \$1,846  | \$1,813  | \$1,864  | \$1,716  | \$2,119  | \$2,010  | \$2,334  | \$2,732  |
| Insurance, Retirement and Other  | \$2,258  | \$2,405  | \$2,595  | \$2,737  | \$2,926  | \$3,148  | \$3,125  | \$3,143  | \$3,905  | \$3,390  | \$3,662  | \$3,769  | \$4,357  | \$4,048  | \$4,506  | \$5,009  |
| Teachers                         | \$5,148  | \$5,382  | \$5,545  | \$5,277  | \$5,430  | \$5,906  | \$6,187  | \$6,222  | \$6,245  | \$6,520  | \$6,792  | \$6,982  | \$8,015  | \$7,858  | \$7,808  | \$8,507  |
| Total                            | \$14,738 | \$15,291 | \$15,950 | \$16,161 | \$16,981 | \$17,882 | \$18,452 | \$18,770 | \$19,700 | \$19,319 | \$20,577 | \$20,900 | \$24,596 | \$24,423 | \$25,334 | \$27,504 |

#### FTE Enrollment

| Table 2                    |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|----------------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| FTE Series                 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
| In-District FTE Pupils     | 131  | 127  | 125  | 125  | 123  | 120  | 122  | 116  | 121  | 123  | 122  | 122  | 107  | 115  | 114  | 113  |
| Foundation Enrollment      | 123  | 120  | 116  | 109  | 109  | 108  | 107  | 111  | 108  | 107  | 105  | 102  | 101  | 91   | 93   | 94   |
| Out-of-District FTE Pupils | 41   | 41   | 41   | 37   | 40   | 38   | 37   | 39   | 38   | 36   | 35   | 33   | 31   | 34   | 34   | 32   |

#### NSS/Ch70 Funding Components (\$/foundation enrollment pupil) — Weighted Averages

| Table 3                    |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
| Ch70 Aid                   | \$3K  | \$3K  | \$3K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$5K  | \$5K  | \$5K  |
| Req NSS (minus Ch70)       | \$6K  | \$6K  | \$6K  | \$7K  | \$7K  | \$7K  | \$7K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$9K  | \$9K  | \$9K  | \$10K |
| Actual NSS (minus Req NSS) | \$5K  | \$4K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  | \$8K  | \$8K  | \$10K | \$10K | \$10K | \$12K | \$12K | \$12K |
| Total NSS                  | \$13K | \$14K | \$15K | \$16K | \$17K | \$17K | \$17K | \$18K | \$20K | \$20K | \$22K | \$22K | \$23K | \$26K | \$26K | \$27K |

#### Cohort Composition (FY2024):

**Tiny (0-200 FTE):** 17 districts (Cohort 1: below Q1) **Member districts:** conway, erving, farmington river reg, florida, hancock, hawlemont, leverett, new salem-wendell, pelham, petersham, richmond, rowe, savoy, shutesbury, sunderland, and 2 others

## Data: All Western MA Traditional Districts: Small (201-800 FTE)

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil) — Weighted Averages

Table 4

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$558    | \$505    | \$540    | \$533    | \$585    | \$515    | \$503    | \$583    | \$539    | \$596    | \$558    | \$512    | \$915    | \$777    | \$1,064  | \$1,150  |
| Guidance, Counseling and Testing | \$392    | \$422    | \$418    | \$456    | \$483    | \$482    | \$501    | \$564    | \$588    | \$600    | \$669    | \$662    | \$807    | \$864    | \$896    | \$961    |
| Administration                   | \$523    | \$515    | \$567    | \$601    | \$648    | \$665    | \$701    | \$739    | \$999    | \$752    | \$788    | \$802    | \$854    | \$942    | \$948    | \$976    |
| Instructional Leadership         | \$811    | \$805    | \$838    | \$824    | \$909    | \$947    | \$967    | \$1,024  | \$1,028  | \$1,042  | \$1,016  | \$1,097  | \$1,188  | \$1,260  | \$1,336  | \$1,463  |
| Operations and Maintenance       | \$1,108  | \$1,056  | \$1,043  | \$1,101  | \$1,198  | \$1,264  | \$1,295  | \$1,338  | \$1,254  | \$1,285  | \$1,473  | \$1,422  | \$1,676  | \$1,878  | \$2,176  | \$2,454  |
| Other Teaching Services          | \$1,061  | \$1,089  | \$1,193  | \$1,151  | \$1,289  | \$1,311  | \$1,353  | \$1,446  | \$1,575  | \$1,718  | \$1,799  | \$1,806  | \$1,779  | \$1,899  | \$2,151  | \$2,402  |
| Pupil Services                   | \$1,421  | \$1,366  | \$1,441  | \$1,537  | \$1,733  | \$1,739  | \$1,714  | \$1,913  | \$1,850  | \$1,870  | \$1,974  | \$2,007  | \$2,097  | \$2,569  | \$2,760  | \$3,122  |
| Insurance, Retirement and Other  | \$2,128  | \$2,107  | \$2,230  | \$2,317  | \$2,486  | \$2,568  | \$2,711  | \$2,955  | \$3,219  | \$3,345  | \$3,362  | \$3,420  | \$3,734  | \$3,801  | \$3,969  | \$4,254  |
| Teachers                         | \$4,522  | \$4,588  | \$4,679  | \$4,870  | \$5,353  | \$5,423  | \$5,557  | \$5,771  | \$6,026  | \$6,075  | \$6,152  | \$6,450  | \$6,922  | \$7,338  | \$7,596  | \$7,838  |
| Total                            | \$12,523 | \$12,453 | \$12,949 | \$13,388 | \$14,684 | \$14,916 | \$15,302 | \$16,333 | \$17,077 | \$17,283 | \$17,791 | \$18,177 | \$19,972 | \$21,330 | \$22,896 | \$24,622 |

### FTE Enrollment

Table 5

| FTE Series                 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
|----------------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| In-District FTE Pupils     | 775  | 779  | 763  | 742  | 696  | 680  | 661  | 644  | 628  | 614  | 617  | 601  | 560  | 556  | 553  | 542  |
| Foundation Enrollment      | 695  | 676  | 666  | 646  | 628  | 603  | 590  | 559  | 542  | 536  | 528  | 526  | 522  | 494  | 484  | 481  |
| Out-of-District FTE Pupils | 75   | 78   | 73   | 73   | 80   | 82   | 86   | 82   | 85   | 88   | 90   | 88   | 85   | 87   | 87   | 84   |

### NSS/Ch70 Funding Components (\$/foundation enrollment pupil) — Weighted Averages

Table 6

| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Ch70 Aid                   | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  |
| Req NSS (minus Ch70)       | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  | \$7K  | \$7K  | \$7K  | \$7K  | \$7K  | \$7K  | \$7K  | \$8K  | \$8K  | \$9K  | \$9K  |
| Actual NSS (minus Req NSS) | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$3K  | \$4K  | \$4K  | \$4K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  |
| Total NSS                  | \$11K | \$11K | \$11K | \$12K | \$13K | \$14K | \$14K | \$15K | \$16K | \$16K | \$17K | \$17K | \$18K | \$19K | \$21K | \$21K |

### Cohort Composition (FY2024):

**Small (201-800 FTE):** 13 districts (Cohort 2: Q1 to median) **Member districts:** clarksburg, deerfield, frontier, gateway, granby, hadley, hatfield, lee, lenox, orange, pioneer valley, ralph c mahar, southern berkshire

## Data: All Western MA Traditional Districts: Medium (801-1600 FTE)

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil) — Weighted Averages

Table 7

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$461    | \$484    | \$558    | \$494    | \$561    | \$552    | \$524    | \$573    | \$586    | \$534    | \$644    | \$607    | \$930    | \$1,064  | \$1,055  | \$1,200  |
| Guidance, Counseling and Testing | \$373    | \$385    | \$391    | \$418    | \$435    | \$457    | \$470    | \$479    | \$522    | \$536    | \$545    | \$601    | \$677    | \$736    | \$805    | \$826    |
| Administration                   | \$494    | \$521    | \$512    | \$533    | \$575    | \$576    | \$614    | \$653    | \$678    | \$711    | \$653    | \$700    | \$747    | \$785    | \$835    | \$911    |
| Instructional Leadership         | \$893    | \$902    | \$885    | \$1,039  | \$1,062  | \$1,091  | \$1,113  | \$1,169  | \$1,150  | \$1,071  | \$1,063  | \$1,113  | \$1,191  | \$1,263  | \$1,373  | \$1,573  |
| Operations and Maintenance       | \$1,150  | \$1,085  | \$1,155  | \$1,095  | \$1,146  | \$1,208  | \$1,233  | \$1,218  | \$1,228  | \$1,203  | \$1,326  | \$1,383  | \$1,564  | \$1,663  | \$1,836  | \$1,953  |
| Other Teaching Services          | \$1,014  | \$1,078  | \$1,100  | \$1,135  | \$1,235  | \$1,317  | \$1,362  | \$1,400  | \$1,444  | \$1,596  | \$1,681  | \$1,729  | \$1,936  | \$2,180  | \$2,311  | \$2,395  |
| Pupil Services                   | \$1,351  | \$1,309  | \$1,366  | \$1,345  | \$1,466  | \$1,527  | \$1,626  | \$1,705  | \$1,686  | \$1,869  | \$1,919  | \$1,803  | \$1,957  | \$2,231  | \$2,548  | \$2,755  |
| Insurance, Retirement and Other  | \$2,483  | \$2,574  | \$2,743  | \$2,707  | \$2,707  | \$2,943  | \$3,074  | \$3,251  | \$3,266  | \$3,622  | \$3,631  | \$3,789  | \$4,169  | \$4,166  | \$4,480  | \$4,740  |
| Teachers                         | \$4,813  | \$4,890  | \$5,055  | \$5,048  | \$5,152  | \$5,357  | \$5,513  | \$5,578  | \$5,674  | \$5,960  | \$6,251  | \$6,385  | \$6,758  | \$6,773  | \$7,182  | \$7,431  |
| Total                            | \$13,032 | \$13,227 | \$13,766 | \$13,815 | \$14,340 | \$15,028 | \$15,529 | \$16,025 | \$16,234 | \$17,103 | \$17,713 | \$18,110 | \$19,930 | \$20,859 | \$22,426 | \$23,785 |

### FTE Enrollment

Table 8

| FTE Series                 | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| In-District FTE Pupils     | 1,466 | 1,434 | 1,416 | 1,402 | 1,388 | 1,334 | 1,306 | 1,277 | 1,268 | 1,247 | 1,268 | 1,249 | 1,160 | 1,175 | 1,158 | 1,144 |
| Foundation Enrollment      | 1,622 | 1,596 | 1,567 | 1,528 | 1,509 | 1,507 | 1,472 | 1,451 | 1,422 | 1,399 | 1,381 | 1,376 | 1,361 | 1,289 | 1,283 | 1,247 |
| Out-of-District FTE Pupils | 154   | 161   | 164   | 164   | 166   | 167   | 176   | 185   | 187   | 194   | 201   | 198   | 190   | 191   | 194   | 194   |

### NSS/Ch70 Funding Components (\$/foundation enrollment pupil) — Weighted Averages

Table 9

| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Ch70 Aid                   | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$7K  | \$7K  | \$7K  | \$8K  | \$8K  |
| Req NSS (minus Ch70)       | \$4K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$7K  | \$7K  | \$8K  |
| Actual NSS (minus Req NSS) | \$2K  | \$1K  | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$4K  | \$4K  |
| Total NSS                  | \$11K | \$11K | \$11K | \$12K | \$12K | \$13K | \$13K | \$14K | \$14K | \$15K | \$16K | \$16K | \$16K | \$17K | \$18K | \$20K |

### Cohort Composition (FY2024):

**Medium (801-1600 FTE):** 15 districts (Cohort 3: median to Q3) **Member districts:** amherst, amherst-pelham, berkshire hills, central berkshire, easthampton, gill-montague, greenfield, hoosac valley regional, mohawk trail, monson, mount greylock, north adams, palmer, southwick-tolland-granville regional school district, ware

## Data: All Western MA Traditional Districts: Large (1601-10K FTE)

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil) — Weighted Averages

Table 10

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$513    | \$648    | \$614    | \$611    | \$654    | \$631    | \$610    | \$644    | \$664    | \$667    | \$722    | \$716    | \$972    | \$1,075  | \$1,331  | \$1,348  |
| Guidance, Counseling and Testing | \$397    | \$394    | \$416    | \$407    | \$423    | \$443    | \$452    | \$454    | \$454    | \$483    | \$511    | \$537    | \$597    | \$664    | \$741    | \$740    |
| Administration                   | \$392    | \$415    | \$426    | \$437    | \$441    | \$462    | \$484    | \$496    | \$480    | \$494    | \$453    | \$502    | \$537    | \$571    | \$616    | \$649    |
| Instructional Leadership         | \$811    | \$859    | \$865    | \$856    | \$837    | \$886    | \$915    | \$932    | \$956    | \$940    | \$998    | \$1,028  | \$1,060  | \$1,153  | \$1,195  | \$1,231  |
| Operations and Maintenance       | \$1,014  | \$993    | \$1,009  | \$986    | \$984    | \$1,023  | \$1,093  | \$1,098  | \$1,111  | \$1,152  | \$1,244  | \$1,238  | \$1,401  | \$1,450  | \$1,685  | \$1,683  |
| Other Teaching Services          | \$990    | \$1,037  | \$1,103  | \$1,105  | \$1,177  | \$1,228  | \$1,290  | \$1,365  | \$1,383  | \$1,395  | \$1,407  | \$1,429  | \$1,561  | \$1,717  | \$1,951  | \$2,021  |
| Pupil Services                   | \$1,135  | \$1,141  | \$1,206  | \$1,205  | \$1,274  | \$1,354  | \$1,396  | \$1,472  | \$1,495  | \$1,586  | \$1,659  | \$1,591  | \$1,603  | \$1,989  | \$2,213  | \$2,408  |
| Insurance, Retirement and Other  | \$1,911  | \$1,943  | \$2,045  | \$2,123  | \$2,115  | \$2,282  | \$2,366  | \$2,419  | \$2,543  | \$2,652  | \$2,773  | \$2,855  | \$3,037  | \$3,212  | \$3,364  | \$3,469  |
| Teachers                         | \$4,613  | \$4,604  | \$4,707  | \$4,868  | \$4,990  | \$5,067  | \$5,194  | \$5,393  | \$5,482  | \$5,569  | \$5,863  | \$6,031  | \$6,461  | \$6,780  | \$7,001  | \$7,292  |
| Total                            | \$11,775 | \$12,034 | \$12,391 | \$12,597 | \$12,895 | \$13,376 | \$13,801 | \$14,273 | \$14,568 | \$14,937 | \$15,629 | \$15,927 | \$17,229 | \$18,612 | \$20,097 | \$20,841 |

### FTE Enrollment

Table 11

| FTE Series                 | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| In-District FTE Pupils     | 4,001 | 3,953 | 3,935 | 3,852 | 3,812 | 3,760 | 3,721 | 3,674 | 3,690 | 3,680 | 3,616 | 3,598 | 3,397 | 3,422 | 3,384 | 3,379 |
| Foundation Enrollment      | 4,432 | 4,382 | 4,347 | 4,325 | 4,280 | 4,230 | 4,191 | 4,163 | 4,093 | 4,077 | 4,037 | 4,013 | 3,975 | 3,823 | 3,787 | 3,752 |
| Out-of-District FTE Pupils | 215   | 227   | 244   | 253   | 256   | 272   | 293   | 309   | 316   | 322   | 341   | 345   | 344   | 341   | 354   | 357   |

### NSS/Ch70 Funding Components (\$/foundation enrollment pupil) — Weighted Averages

Table 12

| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Ch70 Aid                   | \$6K  | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$7K  | \$7K  | \$7K  | \$7K  | \$8K  | \$9K  | \$10K |
| Req NSS (minus Ch70)       | \$3K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  |
| Actual NSS (minus Req NSS) | \$900 | \$708 | \$901 | \$905 | \$944 | \$1K  | \$1K  | \$1K  | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  |
| Total NSS                  | \$10K | \$10K | \$10K | \$11K | \$11K | \$12K | \$12K | \$12K | \$13K | \$13K | \$14K | \$14K | \$15K | \$16K | \$17K | \$18K |

### Cohort Composition (FY2024):

**Large (1601-10K FTE):** 14 districts (Cohort 4: above Q3) **Member districts:** agawam, athol-royalston, belchertown, chicopee, east longmeadow, hampden-wilbraham, holyoke, longmeadow, ludlow, northampton, pittsfield, south hadley, west springfield, westfield

## Data: All Western MA Traditional Districts: Outliers (Springfield at 23,723 FTE in 2024)

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil) — Weighted Averages

Table 13

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$1,143  | \$1,337  | \$2,068  | \$1,143  | \$1,424  | \$1,734  | \$1,241  | \$1,376  | \$1,286  | \$1,371  | \$1,673  | \$1,765  | \$2,487  | \$2,260  | \$2,923  | \$3,071  |
| Guidance, Counseling and Testing | \$380    | \$362    | \$427    | \$388    | \$400    | \$405    | \$408    | \$389    | \$391    | \$438    | \$469    | \$483    | \$513    | \$522    | \$621    | \$697    |
| Administration                   | \$533    | \$654    | \$446    | \$645    | \$550    | \$677    | \$596    | \$580    | \$507    | \$490    | \$416    | \$407    | \$441    | \$501    | \$599    | \$609    |
| Instructional Leadership         | \$719    | \$778    | \$892    | \$1,173  | \$1,257  | \$1,245  | \$1,208  | \$1,282  | \$1,280  | \$1,060  | \$983    | \$1,090  | \$1,103  | \$1,402  | \$1,534  | \$1,571  |
| Operations and Maintenance       | \$1,150  | \$1,081  | \$1,262  | \$1,176  | \$1,132  | \$997    | \$1,121  | \$1,027  | \$1,063  | \$1,156  | \$1,337  | \$1,093  | \$1,534  | \$1,536  | \$3,453  | \$3,806  |
| Other Teaching Services          | \$1,143  | \$1,002  | \$1,146  | \$1,130  | \$1,051  | \$1,086  | \$1,150  | \$1,135  | \$1,070  | \$1,036  | \$1,357  | \$1,390  | \$1,337  | \$1,518  | \$1,776  | \$2,014  |
| Pupil Services                   | \$1,651  | \$1,511  | \$1,729  | \$1,965  | \$1,783  | \$1,852  | \$1,985  | \$2,078  | \$2,222  | \$2,407  | \$2,499  | \$2,148  | \$2,077  | \$3,080  | \$3,054  | \$3,204  |
| Insurance, Retirement and Other  | \$2,050  | \$2,226  | \$2,280  | \$2,474  | \$2,437  | \$2,496  | \$2,467  | \$2,566  | \$2,562  | \$2,635  | \$2,765  | \$3,001  | \$3,217  | \$3,461  | \$3,852  | \$3,906  |
| Teachers                         | \$5,839  | \$4,859  | \$4,750  | \$4,711  | \$4,848  | \$4,823  | \$4,963  | \$5,196  | \$4,962  | \$5,274  | \$5,510  | \$5,800  | \$6,442  | \$7,095  | \$7,561  | \$7,737  |
| Total                            | \$14,608 | \$13,810 | \$15,000 | \$14,805 | \$14,882 | \$15,315 | \$15,139 | \$15,629 | \$15,343 | \$15,867 | \$17,009 | \$17,177 | \$19,151 | \$21,375 | \$25,373 | \$26,615 |

### FTE Enrollment

Table 14

| FTE Series                 | 2009   | 2010   | 2011   | 2012   | 2013   | 2014   | 2015   | 2016   | 2017   | 2018   | 2019   | 2020   | 2021   | 2022   | 2023   | 2024   |
|----------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| In-District FTE Pupils     | 25,137 | 24,987 | 24,935 | 24,521 | 24,747 | 25,249 | 25,101 | 25,009 | 25,629 | 25,776 | 25,294 | 24,906 | 24,079 | 24,040 | 23,763 | 23,723 |
| Foundation Enrollment      | 28,235 | 28,343 | 28,305 | 28,226 | 27,951 | 28,434 | 28,755 | 28,970 | 29,109 | 29,533 | 29,670 | 29,645 | 29,551 | 29,144 | 28,613 | 28,326 |
| Out-of-District FTE Pupils | 3,571  | 3,775  | 3,719  | 3,695  | 3,817  | 3,613  | 3,974  | 4,406  | 4,596  | 4,812  | 5,150  | 5,363  | 5,559  | 5,500  | 5,652  | 5,745  |

### NSS/Ch70 Funding Components (\$/foundation enrollment pupil) — Weighted Averages

Table 15

| Component                  | 2009   | 2010   | 2011  | 2012  | 2013   | 2014  | 2015  | 2016  | 2017  | 2018   | 2019  | 2020   | 2021  | 2022  | 2023  | 2024  |
|----------------------------|--------|--------|-------|-------|--------|-------|-------|-------|-------|--------|-------|--------|-------|-------|-------|-------|
| Ch70 Aid                   | \$9K   | \$9K   | \$9K  | \$10K | \$10K  | \$10K | \$10K | \$11K | \$11K | \$11K  | \$12K | \$13K  | \$13K | \$14K | \$15K | \$17K |
| Req NSS (minus Ch70)       | \$247  | \$1K   | \$2K  | \$1K  | \$1K   | \$1K  | \$1K  | \$1K  | \$1K  | \$1K   | \$1K  | \$1K   | \$2K  | \$1K  | \$2K  | \$2K  |
| Actual NSS (minus Req NSS) | \$-150 | \$-384 | \$-35 | \$-84 | \$-144 | \$0   | \$2   | \$2   | \$-0  | \$-106 | \$-0  | \$-499 | \$0   | \$0   | \$-0  | \$-0  |
| Total NSS                  | \$9K   | \$10K  | \$11K | \$11K | \$11K  | \$12K | \$12K | \$12K | \$12K | \$12K  | \$13K | \$13K  | \$15K | \$15K | \$17K | \$19K |

### Cohort Composition (FY2024):

**Outliers (Springfield >10K FTE):** 1 district (Cohort 5: statistical outlier, analyzed separately) **Member districts:** springfield



## Data: All Western MA Traditional Districts: All Western MA (excl. Springfield)

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil)

Table 16

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$507    | \$590    | \$593    | \$575    | \$624    | \$600    | \$580    | \$622    | \$636    | \$627    | \$686    | \$671    | \$967    | \$1,043  | \$1,240  | \$1,287  |
| Guidance, Counseling and Testing | \$389    | \$394    | \$409    | \$415    | \$432    | \$451    | \$462    | \$471    | \$483    | \$506    | \$534    | \$564    | \$636    | \$700    | \$768    | \$780    |
| Administration                   | \$440    | \$461    | \$471    | \$488    | \$507    | \$520    | \$544    | \$566    | \$589    | \$580    | \$542    | \$587    | \$628    | \$669    | \$713    | \$754    |
| Instructional Leadership         | \$836    | \$869    | \$873    | \$906    | \$910    | \$951    | \$981    | \$1,013  | \$1,023  | \$996    | \$1,029  | \$1,070  | \$1,122  | \$1,205  | \$1,266  | \$1,353  |
| Operations and Maintenance       | \$1,066  | \$1,030  | \$1,058  | \$1,035  | \$1,055  | \$1,105  | \$1,158  | \$1,160  | \$1,161  | \$1,185  | \$1,301  | \$1,302  | \$1,482  | \$1,566  | \$1,783  | \$1,845  |
| Other Teaching Services          | \$1,025  | \$1,073  | \$1,133  | \$1,141  | \$1,232  | \$1,284  | \$1,342  | \$1,413  | \$1,449  | \$1,508  | \$1,553  | \$1,582  | \$1,714  | \$1,895  | \$2,100  | \$2,190  |
| Pupil Services                   | \$1,224  | \$1,214  | \$1,276  | \$1,283  | \$1,377  | \$1,442  | \$1,491  | \$1,580  | \$1,584  | \$1,686  | \$1,758  | \$1,685  | \$1,747  | \$2,103  | \$2,348  | \$2,566  |
| Insurance, Retirement and Other  | \$2,083  | \$2,124  | \$2,247  | \$2,301  | \$2,318  | \$2,491  | \$2,589  | \$2,689  | \$2,816  | \$2,966  | \$3,060  | \$3,157  | \$3,404  | \$3,516  | \$3,715  | \$3,881  |
| Teachers                         | \$4,664  | \$4,689  | \$4,806  | \$4,921  | \$5,078  | \$5,194  | \$5,331  | \$5,496  | \$5,601  | \$5,735  | \$6,009  | \$6,181  | \$6,614  | \$6,861  | \$7,123  | \$7,409  |
| Total                            | \$12,234 | \$12,444 | \$12,865 | \$13,063 | \$13,532 | \$14,039 | \$14,479 | \$15,010 | \$15,343 | \$15,789 | \$16,472 | \$16,798 | \$18,314 | \$19,558 | \$21,055 | \$22,065 |

### FTE Enrollment

Table 17

| FTE Series                 | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| In-District FTE Pupils     | 1,555 | 1,534 | 1,521 | 1,493 | 1,469 | 1,438 | 1,418 | 1,372 | 1,371 | 1,361 | 1,352 | 1,339 | 1,255 | 1,266 | 1,252 | 1,245 |
| Foundation Enrollment      | 1,603 | 1,580 | 1,563 | 1,543 | 1,524 | 1,505 | 1,486 | 1,469 | 1,442 | 1,432 | 1,416 | 1,408 | 1,394 | 1,333 | 1,322 | 1,306 |
| Out-of-District FTE Pupils | 120   | 125   | 129   | 130   | 134   | 138   | 146   | 150   | 152   | 155   | 162   | 161   | 158   | 158   | 162   | 162   |

### NSS/Ch70 Funding Components (\$/foundation enrollment pupil)

Table 18

| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Ch70 Aid                   | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$7K  | \$7K  | \$8K  | \$8K  | \$9K  |
| Req NSS (minus Ch70)       | \$4K  | \$4K  | \$4K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  | \$7K  |
| Actual NSS (minus Req NSS) | \$1K  | \$1K  | \$1K  | \$1K  | \$1K  | \$1K  | \$2K  | \$2K  | \$2K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  |
| Total NSS                  | \$10K | \$11K | \$11K | \$11K | \$12K | \$12K | \$13K | \$13K | \$14K | \$14K | \$15K | \$15K | \$15K | \$17K | \$18K | \$19K |

## Data: Amherst-Pelham

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil)

Table 19

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$646    | \$534    | \$900    | \$417    | \$459    | \$424    | \$599    | \$860    | \$635    | \$384    | \$610    | \$559    | \$477    | \$527    | \$457    | \$799    |
| Guidance, Counseling and Testing | \$525    | \$580    | \$617    | \$684    | \$652    | \$733    | \$778    | \$798    | \$803    | \$891    | \$925    | \$945    | \$1,001  | \$1,241  | \$1,406  | \$1,465  |
| Administration                   | \$652    | \$664    | \$687    | \$789    | \$987    | \$907    | \$953    | \$926    | \$996    | \$857    | \$759    | \$780    | \$788    | \$859    | \$928    | \$1,277  |
| Instructional Leadership         | \$1,289  | \$1,337  | \$1,209  | \$1,397  | \$1,390  | \$1,545  | \$1,429  | \$1,437  | \$1,541  | \$1,555  | \$1,544  | \$1,501  | \$1,436  | \$1,472  | \$1,644  | \$2,336  |
| Operations and Maintenance       | \$1,459  | \$1,416  | \$1,559  | \$1,468  | \$1,481  | \$1,585  | \$1,473  | \$1,577  | \$1,904  | \$1,678  | \$1,776  | \$1,601  | \$1,705  | \$1,778  | \$1,957  | \$2,221  |
| Other Teaching Services          | \$1,199  | \$1,264  | \$1,313  | \$1,468  | \$1,588  | \$1,739  | \$1,770  | \$1,883  | \$2,014  | \$2,066  | \$2,122  | \$2,214  | \$3,010  | \$3,050  | \$2,968  | \$2,851  |
| Pupil Services                   | \$1,645  | \$1,616  | \$1,588  | \$1,618  | \$1,639  | \$1,764  | \$1,975  | \$2,110  | \$2,128  | \$2,218  | \$2,236  | \$1,957  | \$2,950  | \$2,539  | \$2,771  | \$2,766  |
| Insurance, Retirement and Other  | \$3,305  | \$3,636  | \$3,673  | \$3,811  | \$3,526  | \$3,954  | \$4,086  | \$4,103  | \$4,218  | \$5,006  | \$5,047  | \$5,074  | \$5,237  | \$5,062  | \$5,887  | \$6,248  |
| Teachers                         | \$5,491  | \$5,511  | \$6,254  | \$6,199  | \$6,533  | \$6,765  | \$6,704  | \$6,590  | \$6,927  | \$7,208  | \$7,285  | \$7,155  | \$7,592  | \$7,404  | \$8,158  | \$8,270  |
| Total                            | \$16,211 | \$16,558 | \$17,800 | \$17,851 | \$18,255 | \$19,416 | \$19,767 | \$20,284 | \$21,166 | \$21,863 | \$22,304 | \$21,786 | \$24,196 | \$23,932 | \$26,176 | \$28,233 |

### FTE Enrollment

Table 20

| FTE Series                 | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| In-District FTE Pupils     | 1,712 | 1,659 | 1,573 | 1,536 | 1,519 | 1,470 | 1,450 | 1,378 | 1,366 | 1,342 | 1,334 | 1,346 | 1,278 | 1,265 | 1,241 | 1,209 |
| Foundation Enrollment      | 1,801 | 1,763 | 1,726 | 1,647 | 1,610 | 1,581 | 1,529 | 1,532 | 1,496 | 1,490 | 1,433 | 1,399 | 1,399 | 1,330 | 1,310 | 1,274 |
| Out-of-District FTE Pupils | 112   | 129   | 138   | 143   | 138   | 149   | 169   | 190   | 192   | 170   | 158   | 152   | 155   | 145   | 150   | 147   |

### NSS/Ch70 Funding Components (\$/foundation enrollment pupil)

Table 21

| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Ch70 Aid                   | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$7K  | \$7K  | \$7K  | \$7K  | \$7K  | \$8K  |
| Req NSS (minus Ch70)       | \$4K  | \$6K  | \$6K  | \$7K  | \$7K  | \$7K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$9K  | \$9K  | \$10K | \$10K | \$11K |
| Actual NSS (minus Req NSS) | \$3K  | \$2K  | \$3K  | \$3K  | \$4K  | \$4K  | \$4K  | \$4K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  |
| Total NSS                  | \$13K | \$14K | \$14K | \$16K | \$17K | \$17K | \$18K | \$18K | \$19K | \$19K | \$20K | \$21K | \$21K | \$22K | \$23K | \$25K |

## Data: Amherst

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil)

Table 22

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$684    | \$547    | \$606    | \$931    | \$1,098  | \$1,098  | \$1,265  | \$1,242  | \$1,119  | \$798    | \$1,092  | \$733    | \$1,563  | \$1,561  | \$1,475  | \$1,328  |
| Guidance, Counseling and Testing | \$396    | \$383    | \$362    | \$411    | \$469    | \$538    | \$545    | \$583    | \$599    | \$631    | \$668    | \$696    | \$830    | \$891    | \$1,093  | \$1,053  |
| Administration                   | \$627    | \$698    | \$735    | \$709    | \$787    | \$809    | \$919    | \$944    | \$981    | \$799    | \$767    | \$827    | \$874    | \$896    | \$981    | \$1,213  |
| Instructional Leadership         | \$1,134  | \$1,079  | \$965    | \$1,032  | \$1,153  | \$1,275  | \$1,175  | \$1,131  | \$1,087  | \$1,231  | \$1,325  | \$1,527  | \$1,750  | \$1,783  | \$1,847  | \$2,498  |
| Operations and Maintenance       | \$1,264  | \$1,271  | \$1,636  | \$1,436  | \$1,399  | \$1,451  | \$1,577  | \$1,530  | \$1,555  | \$1,340  | \$1,646  | \$1,764  | \$1,897  | \$1,874  | \$1,941  | \$2,248  |
| Other Teaching Services          | \$1,762  | \$1,813  | \$1,923  | \$2,345  | \$2,621  | \$2,610  | \$2,840  | \$2,652  | \$2,751  | \$2,973  | \$3,084  | \$3,053  | \$3,908  | \$3,898  | \$3,746  | \$4,612  |
| Pupil Services                   | \$865    | \$1,003  | \$973    | \$975    | \$1,084  | \$1,223  | \$1,257  | \$1,267  | \$1,219  | \$1,278  | \$1,362  | \$1,148  | \$1,205  | \$1,849  | \$1,728  | \$1,965  |
| Insurance, Retirement and Other  | \$3,353  | \$3,610  | \$3,821  | \$3,852  | \$3,941  | \$4,189  | \$4,201  | \$4,295  | \$4,231  | \$4,840  | \$5,212  | \$5,363  | \$5,661  | \$5,561  | \$6,022  | \$6,605  |
| Teachers                         | \$5,944  | \$6,213  | \$6,355  | \$6,963  | \$6,804  | \$6,627  | \$6,841  | \$7,712  | \$7,792  | \$8,091  | \$8,538  | \$8,725  | \$9,528  | \$9,598  | \$9,451  | \$9,745  |
| Total                            | \$16,029 | \$16,617 | \$17,376 | \$18,654 | \$19,356 | \$19,820 | \$20,620 | \$21,356 | \$21,334 | \$21,981 | \$23,694 | \$23,836 | \$27,216 | \$27,911 | \$28,284 | \$31,267 |

### FTE Enrollment

Table 23

| FTE Series                 | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| In-District FTE Pupils     | 1,387 | 1,331 | 1,284 | 1,215 | 1,196 | 1,197 | 1,184 | 1,154 | 1,156 | 1,163 | 1,131 | 1,107 | 1,017 | 1,052 | 1,062 | 1,002 |
| Foundation Enrollment      | 1,398 | 1,405 | 1,356 | 1,279 | 1,269 | 1,227 | 1,205 | 1,192 | 1,199 | 1,136 | 1,106 | 1,134 | 1,097 | 1,045 | 1,065 | 1,055 |
| Out-of-District FTE Pupils | 65    | 72    | 73    | 84    | 90    | 90    | 101   | 113   | 110   | 108   | 124   | 122   | 118   | 123   | 129   | 134   |

### NSS/Ch70 Funding Components (\$/foundation enrollment pupil)

Table 24

| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Ch70 Aid                   | \$4K  | \$4K  | \$4K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  |
| Req NSS (minus Ch70)       | \$4K  | \$6K  | \$6K  | \$7K  | \$7K  | \$7K  | \$7K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$9K  | \$9K  | \$10K | \$11K |
| Actual NSS (minus Req NSS) | \$6K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  | \$7K  | \$8K  | \$8K  | \$9K  | \$10K | \$9K  | \$9K  | \$11K | \$11K | \$11K |
| Total NSS                  | \$15K | \$15K | \$16K | \$17K | \$17K | \$18K | \$19K | \$20K | \$20K | \$22K | \$23K | \$23K | \$24K | \$26K | \$26K | \$28K |

## Data: Leverett

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil)

Table 25

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$575    | \$602    | \$651    | \$766    | \$616    | \$556    | \$563    | \$529    | \$596    | \$453    | \$503    | \$606    | \$1,041  | \$840    | \$756    | \$608    |
| Guidance, Counseling and Testing | \$310    | \$306    | \$314    | \$350    | \$354    | \$412    | \$292    | \$304    | \$323    | \$355    | \$351    | \$362    | \$498    | \$790    | \$659    | \$666    |
| Administration                   | \$715    | \$676    | \$710    | \$888    | \$925    | \$1,029  | \$809    | \$842    | \$1,244  | \$899    | \$826    | \$857    | \$1,010  | \$987    | \$1,061  | \$953    |
| Instructional Leadership         | \$1,015  | \$1,008  | \$1,072  | \$1,244  | \$1,257  | \$1,393  | \$1,357  | \$1,344  | \$1,431  | \$1,461  | \$1,278  | \$1,317  | \$1,318  | \$1,354  | \$1,430  | \$1,663  |
| Operations and Maintenance       | \$1,755  | \$1,427  | \$1,532  | \$1,741  | \$1,479  | \$2,421  | \$2,272  | \$1,707  | \$1,209  | \$1,765  | \$1,491  | \$1,458  | \$2,131  | \$2,296  | \$2,292  | \$2,158  |
| Other Teaching Services          | \$2,385  | \$2,040  | \$2,074  | \$2,346  | \$2,540  | \$2,352  | \$2,274  | \$2,281  | \$2,193  | \$2,175  | \$2,477  | \$2,386  | \$2,178  | \$3,119  | \$3,266  | \$3,526  |
| Pupil Services                   | \$1,085  | \$664    | \$782    | \$1,249  | \$1,259  | \$1,980  | \$1,781  | \$1,397  | \$1,759  | \$1,726  | \$1,494  | \$1,279  | \$2,192  | \$1,120  | \$1,976  | \$2,148  |
| Insurance, Retirement and Other  | \$2,098  | \$2,041  | \$2,253  | \$2,499  | \$2,753  | \$3,245  | \$3,041  | \$2,937  | \$2,943  | \$3,677  | \$3,781  | \$3,686  | \$4,766  | \$3,630  | \$3,611  | \$3,793  |
| Teachers                         | \$5,218  | \$5,690  | \$5,818  | \$6,301  | \$6,138  | \$7,547  | \$6,868  | \$6,967  | \$6,814  | \$7,967  | \$7,665  | \$7,134  | \$9,224  | \$6,888  | \$6,724  | \$6,395  |
| Total                            | \$15,156 | \$14,454 | \$15,206 | \$17,384 | \$17,321 | \$20,935 | \$19,257 | \$18,308 | \$18,512 | \$20,478 | \$19,866 | \$19,085 | \$24,358 | \$21,024 | \$21,775 | \$21,910 |

### FTE Enrollment

Table 26

| FTE Series                 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
|----------------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| In-District FTE Pupils     | 162  | 164  | 161  | 150  | 147  | 124  | 134  | 135  | 134  | 129  | 136  | 142  | 117  | 142  | 140  | 142  |
| Foundation Enrollment      | 125  | 120  | 123  | 123  | 119  | 118  | 100  | 104  | 101  | 123  | 103  | 112  | 116  | 105  | 112  | 101  |
| Out-of-District FTE Pupils | 7    | 7    | 6    | 9    | 10   | 9    | 8    | 6    | 14   | 8    | 9    | 12   | 9    | 12   | 7    | 3    |

### NSS/Ch70 Funding Components (\$/foundation enrollment pupil)

Table 27

| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Ch70 Aid                   | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$3K  | \$3K  | \$3K  | \$2K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$4K  |
| Req NSS (minus Ch70)       | \$7K  | \$7K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$9K  | \$9K  | \$9K  | \$9K  |
| Actual NSS (minus Req NSS) | \$7K  | \$7K  | \$7K  | \$8K  | \$9K  | \$8K  | \$10K | \$10K | \$10K | \$8K  | \$11K | \$11K | \$10K | \$13K | \$10K | \$11K |
| Total NSS                  | \$16K | \$17K | \$17K | \$17K | \$19K | \$18K | \$21K | \$21K | \$21K | \$18K | \$23K | \$22K | \$21K | \$25K | \$22K | \$24K |

## Data: Pelham

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil)

Table 28

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$580    | \$361    | \$384    | \$366    | \$658    | \$639    | \$780    | \$652    | \$569    | \$414    | \$466    | \$407    | \$292    | \$365    | \$422    | \$207    |
| Guidance, Counseling and Testing | \$438    | \$449    | \$459    | \$449    | \$617    | \$664    | \$696    | \$683    | \$682    | \$499    | \$499    | \$582    | \$795    | \$772    | \$775    | \$1,855  |
| Administration                   | \$390    | \$398    | \$478    | \$391    | \$664    | \$796    | \$806    | \$980    | \$739    | \$731    | \$489    | \$574    | \$656    | \$688    | \$614    | \$917    |
| Instructional Leadership         | \$1,123  | \$1,159  | \$1,089  | \$1,064  | \$1,113  | \$1,252  | \$1,288  | \$1,322  | \$1,278  | \$1,431  | \$1,194  | \$1,609  | \$1,670  | \$1,873  | \$1,782  | \$1,885  |
| Operations and Maintenance       | \$920    | \$847    | \$925    | \$884    | \$992    | \$947    | \$1,080  | \$942    | \$987    | \$889    | \$1,229  | \$1,304  | \$1,498  | \$1,380  | \$1,524  | \$1,614  |
| Other Teaching Services          | \$2,184  | \$2,109  | \$2,166  | \$2,026  | \$1,937  | \$2,063  | \$2,218  | \$2,541  | \$2,850  | \$2,269  | \$2,273  | \$2,324  | \$2,540  | \$2,292  | \$2,594  | \$3,010  |
| Pupil Services                   | \$594    | \$641    | \$888    | \$1,058  | \$967    | \$951    | \$1,000  | \$1,114  | \$1,092  | \$1,173  | \$1,233  | \$1,060  | \$1,317  | \$1,431  | \$1,113  | \$1,755  |
| Insurance, Retirement and Other  | \$2,615  | \$2,585  | \$2,602  | \$2,897  | \$3,052  | \$3,283  | \$3,376  | \$3,473  | \$3,863  | \$4,561  | \$3,791  | \$4,168  | \$4,850  | \$4,157  | \$3,983  | \$4,670  |
| Teachers                         | \$5,889  | \$5,474  | \$5,627  | \$5,576  | \$5,035  | \$6,090  | \$6,540  | \$6,418  | \$5,904  | \$6,310  | \$6,129  | \$6,860  | \$7,406  | \$6,714  | \$6,430  | \$7,127  |
| Total                            | \$14,733 | \$14,023 | \$14,618 | \$14,711 | \$15,035 | \$16,685 | \$17,784 | \$18,125 | \$17,964 | \$18,277 | \$17,303 | \$18,888 | \$21,024 | \$19,672 | \$19,237 | \$23,040 |

### FTE Enrollment

Table 29

| FTE Series                 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
|----------------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| In-District FTE Pupils     | 118  | 125  | 122  | 128  | 132  | 128  | 123  | 123  | 130  | 127  | 134  | 125  | 106  | 119  | 134  | 126  |
| Foundation Enrollment      | 85   | 81   | 81   | 68   | 63   | 67   | 70   | 73   | 74   | 81   | 83   | 82   | 84   | 70   | 65   | 63   |
| Out-of-District FTE Pupils | 3    | 3    | 4    | 1    | 4    | 7    | 5    | 7    | 11   | 12   | 10   | 4    | 8    | 6    | 5    | 9    |

### NSS/Ch70 Funding Components (\$/foundation enrollment pupil)

Table 30

| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Ch70 Aid                   | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$4K  | \$4K  |
| Req NSS (minus Ch70)       | \$5K  | \$6K  | \$6K  | \$7K  | \$8K  | \$8K  | \$9K  | \$9K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$9K  | \$10K | \$10K |
| Actual NSS (minus Req NSS) | \$8K  | \$8K  | \$8K  | \$10K | \$12K | \$10K | \$9K  | \$10K | \$12K | \$12K | \$11K | \$11K | \$13K | \$16K | \$19K | \$20K |
| Total NSS                  | \$16K | \$17K | \$17K | \$20K | \$23K | \$22K | \$21K | \$22K | \$23K | \$23K | \$22K | \$23K | \$24K | \$29K | \$32K | \$34K |

## Data: Shutesbury

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil)

Table 31

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$580    | \$498    | \$564    | \$865    | \$704    | \$515    | \$636    | \$684    | \$1,510  | \$420    | \$385    | \$873    | \$1,296  | \$551    | \$1,102  | \$545    |
| Guidance, Counseling and Testing | \$302    | \$328    | \$345    | \$334    | \$345    | \$344    | \$354    | \$434    | \$318    | \$546    | \$542    | \$383    | \$519    | \$1,393  | \$697    | \$826    |
| Administration                   | \$652    | \$943    | \$767    | \$836    | \$877    | \$726    | \$884    | \$1,084  | \$1,040  | \$846    | \$856    | \$938    | \$925    | \$1,100  | \$934    | \$1,015  |
| Instructional Leadership         | \$969    | \$1,138  | \$1,149  | \$1,198  | \$1,227  | \$1,003  | \$1,192  | \$1,517  | \$1,604  | \$1,481  | \$1,517  | \$1,541  | \$1,707  | \$1,736  | \$1,323  | \$2,103  |
| Operations and Maintenance       | \$1,180  | \$1,079  | \$1,221  | \$1,276  | \$1,251  | \$1,338  | \$1,384  | \$1,637  | \$1,791  | \$1,941  | \$1,663  | \$1,916  | \$2,665  | \$2,048  | \$2,051  | \$2,754  |
| Other Teaching Services          | \$2,185  | \$2,447  | \$2,361  | \$2,193  | \$2,179  | \$2,591  | \$2,164  | \$2,817  | \$3,301  | \$3,491  | \$3,545  | \$3,560  | \$3,722  | \$5,320  | \$3,441  | \$4,359  |
| Pupil Services                   | \$990    | \$800    | \$850    | \$1,222  | \$1,283  | \$1,074  | \$1,073  | \$906    | \$1,847  | \$1,755  | \$1,664  | \$1,815  | \$1,937  | \$1,355  | \$2,249  | \$2,136  |
| Insurance, Retirement and Other  | \$2,332  | \$2,687  | \$2,596  | \$2,690  | \$2,564  | \$2,484  | \$2,328  | \$3,107  | \$3,773  | \$4,388  | \$2,970  | \$3,963  | \$3,350  | \$3,154  | \$4,007  | \$3,501  |
| Teachers                         | \$4,821  | \$5,363  | \$5,603  | \$5,169  | \$5,509  | \$6,012  | \$5,620  | \$6,612  | \$7,043  | \$7,838  | \$7,914  | \$8,204  | \$8,933  | \$10,075 | \$8,898  | \$9,669  |
| Total                            | \$14,011 | \$15,283 | \$15,456 | \$15,783 | \$15,939 | \$16,087 | \$15,635 | \$18,798 | \$22,227 | \$22,706 | \$21,056 | \$23,193 | \$25,054 | \$26,732 | \$24,702 | \$26,908 |

### FTE Enrollment

Table 32

| FTE Series                 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
|----------------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| In-District FTE Pupils     | 162  | 151  | 146  | 151  | 150  | 157  | 156  | 131  | 120  | 124  | 127  | 125  | 113  | 115  | 123  | 110  |
| Foundation Enrollment      | 156  | 149  | 141  | 138  | 143  | 132  | 141  | 146  | 118  | 116  | 118  | 117  | 105  | 98   | 100  | 111  |
| Out-of-District FTE Pupils | 7    | 5    | 7    | 10   | 9    | 13   | 9    | 12   | 13   | 16   | 18   | 15   | 16   | 18   | 19   | 18   |

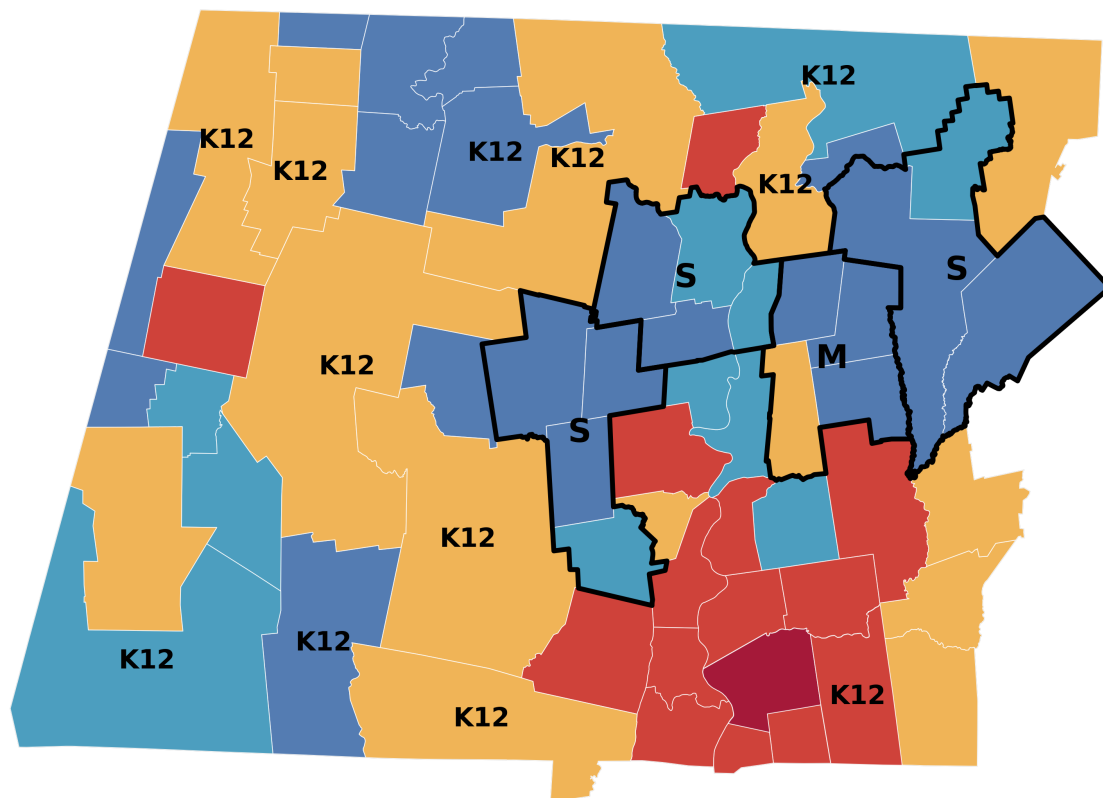
### NSS/Ch70 Funding Components (\$/foundation enrollment pupil)

Table 33

| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Ch70 Aid                   | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  |
| Req NSS (minus Ch70)       | \$4K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  | \$7K  | \$7K  | \$7K  | \$7K  |
| Actual NSS (minus Req NSS) | \$4K  | \$5K  | \$6K  | \$6K  | \$6K  | \$7K  | \$7K  | \$6K  | \$9K  | \$10K | \$10K | \$12K | \$12K | \$15K | \$13K | \$10K |
| Total NSS                  | \$13K | \$14K | \$15K | \$15K | \$15K | \$17K | \$16K | \$15K | \$20K | \$22K | \$21K | \$23K | \$25K | \$29K | \$26K | \$24K |

## Appendix D. Maps from previous years

Geographic map showing district locations and enrollment cohorts (2019)



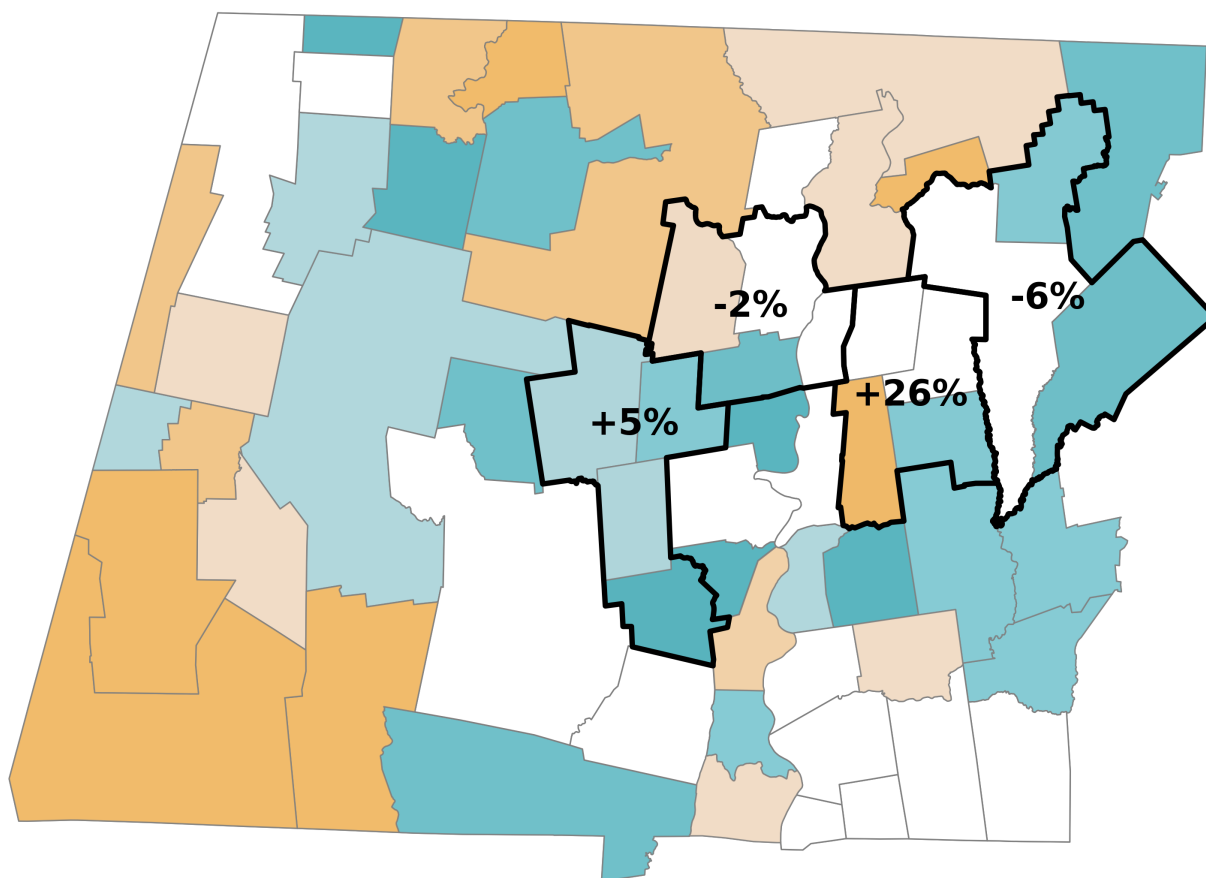
Enrollment Cohorts - 2019

|                                                                                       |                                                                                                                                    |
|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <span style="color: blue;">■</span> Tiny (0-200 FTE): 20 district(s)                  | <span style="border: 2px solid black; display: inline-block; width: 20px; height: 10px;"></span> Secondary regional (black border) |
| <span style="color: lightblue;">■</span> Small (201-800 FTE): 14 district(s)          | Cohort: T, S, M, L, XL                                                                                                             |
| <span style="color: orange;">■</span> Medium (801-1600 FTE): 16 district(s)           | K12 = Regional K-12 with all grade levels in a single district                                                                     |
| <span style="color: red;">■</span> Large (1601-10K FTE): 14 district(s)               |                                                                                                                                    |
| <span style="color: darkred;">■</span> Outliers (Springfield >10K FTE): 1 district(s) |                                                                                                                                    |

Figure 1

## Appendix D. Maps from previous years (continued)

Geographic map showing 2019 PPE vs enrollment cohort baseline



### Total PPE vs Cohort Baseline - 2019 ( $\pm 5\%$ threshold)

|                                                                                                                                                                                         |                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: inline-block; width: 15px; height: 15px; background-color: #008080; border: 1px solid black; margin-right: 5px;"></div> $\leq -20\%$ cohort avg: 6 district(s)     | <div style="display: inline-block; width: 15px; height: 15px; background-color: #F5DEB3; border: 1px solid black; margin-right: 5px;"></div> $+5$ to $+10\%$ cohort avg: 8 district(s)  |
| <div style="display: inline-block; width: 15px; height: 15px; background-color: #4682B4; border: 1px solid black; margin-right: 5px;"></div> $-15$ to $-20\%$ cohort avg: 6 district(s) | <div style="display: inline-block; width: 15px; height: 15px; background-color: #FFDAB9; border: 1px solid black; margin-right: 5px;"></div> $+10$ to $+15\%$ cohort avg: 1 district(s) |
| <div style="display: inline-block; width: 15px; height: 15px; background-color: #6495ED; border: 1px solid black; margin-right: 5px;"></div> $-10$ to $-15\%$ cohort avg: 7 district(s) | <div style="display: inline-block; width: 15px; height: 15px; background-color: #FFA07A; border: 1px solid black; margin-right: 5px;"></div> $+15$ to $+20\%$ cohort avg: 4 district(s) |
| <div style="display: inline-block; width: 15px; height: 15px; background-color: #ADD8E6; border: 1px solid black; margin-right: 5px;"></div> $-5$ to $-10\%$ cohort avg: 7 district(s)  | <div style="display: inline-block; width: 15px; height: 15px; background-color: #FF8C00; border: 1px solid black; margin-right: 5px;"></div> $\geq +20\%$ cohort avg: 7 district(s)     |
| <div style="display: inline-block; width: 15px; height: 15px; background-color: white; border: 1px solid black; margin-right: 5px;"></div> Within $\pm 5\%$ cohort avg: 19 district(s)  | $+/-\%$ = Secondary regional district deviation (n=4)                                                                                                                                   |

Figure 2



## Appendix D. Maps from previous years (continued)

Geographic map showing 10-year PPE growth (2009-2019) vs enrollment cohort baseline

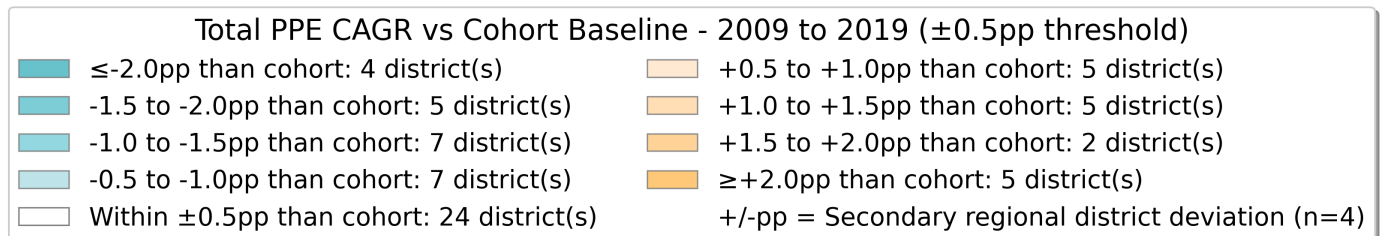
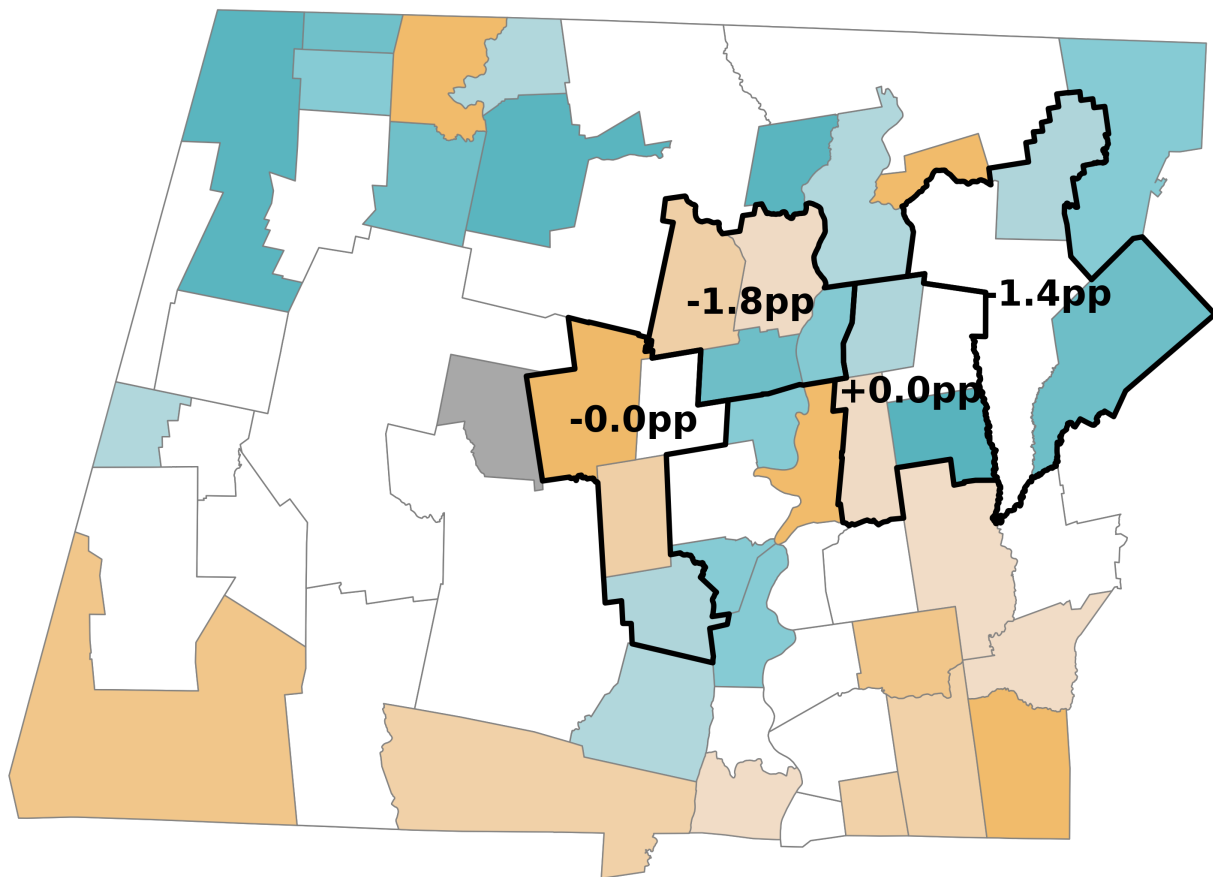


Figure 3

## Appendix D. Maps from previous years (continued)

Geographic map showing district locations and enrollment cohorts (2014)

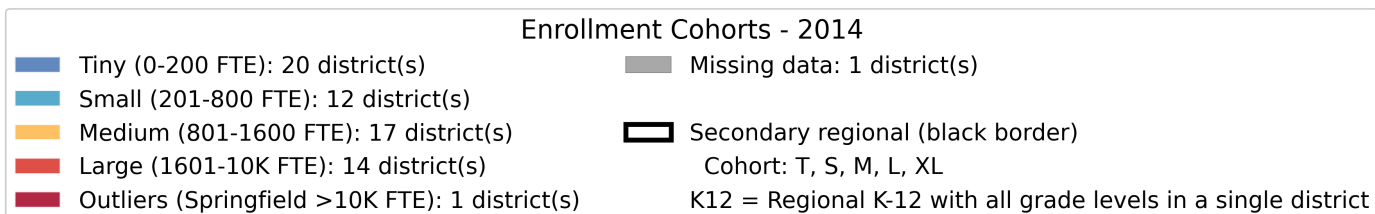
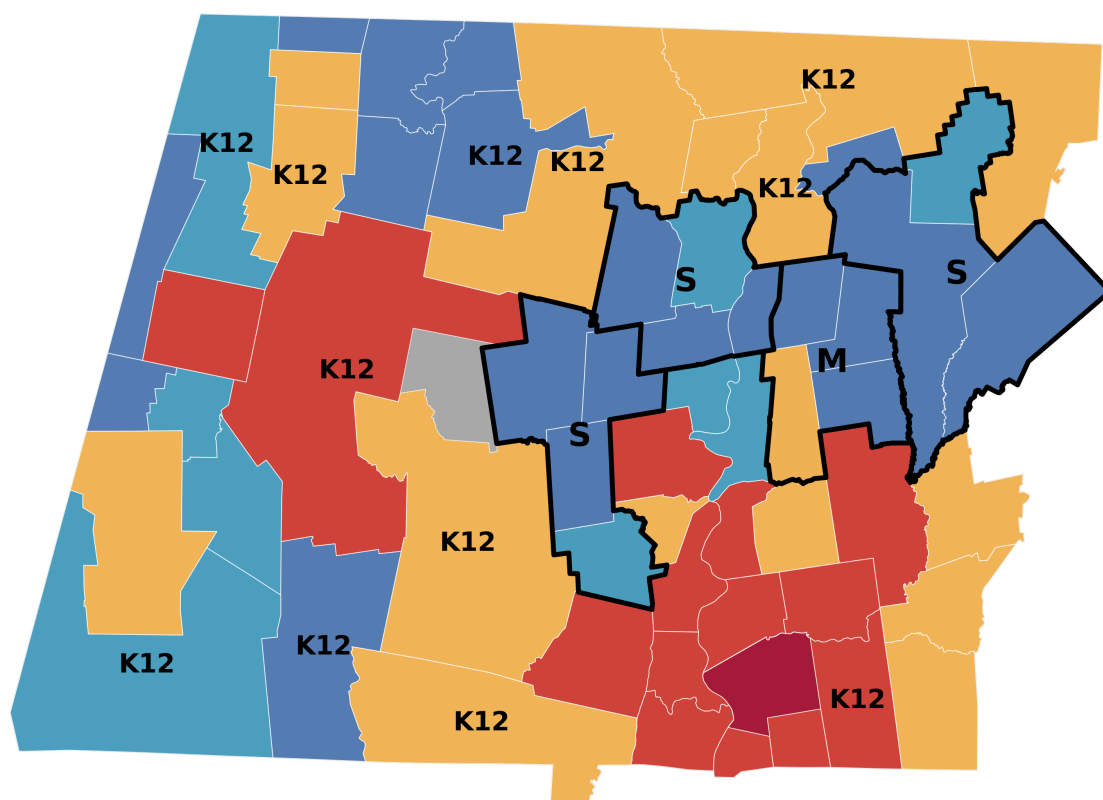
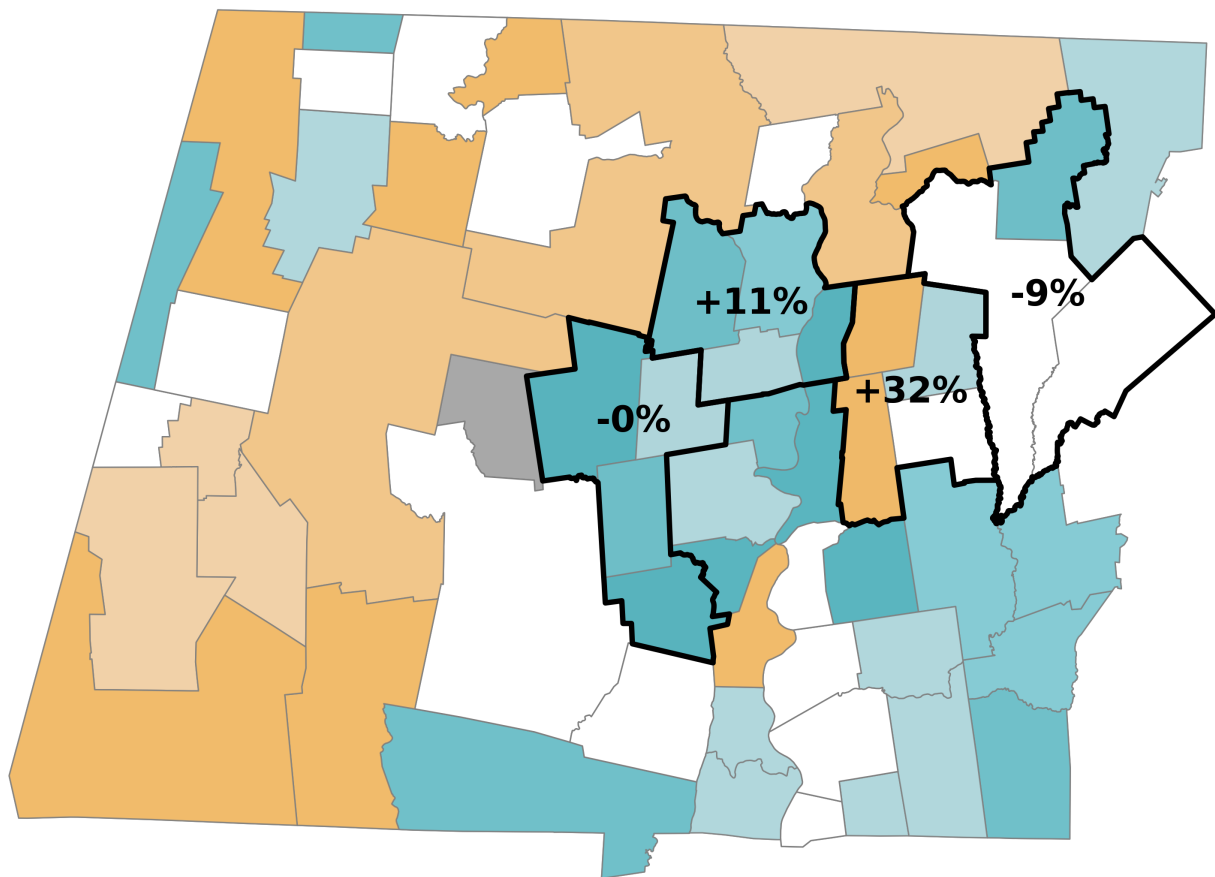


Figure 4

## Appendix D. Maps from previous years (continued)

Geographic map showing 2014 PPE vs enrollment cohort baseline



### Total PPE vs Cohort Baseline - 2014 ( $\pm 5\%$ threshold)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <div style="display: flex; flex-wrap: wrap;"> <div style="width: 50%;"> <div><span style="display: inline-block; width: 15px; height: 10px; background-color: #00728f; margin-right: 5px;"></span> <math>\leq -20\%</math> cohort avg: 6 district(s)</div> <div><span style="display: inline-block; width: 15px; height: 10px; background-color: #0099cc; margin-right: 5px;"></span> -15 to -20% cohort avg: 8 district(s)</div> <div><span style="display: inline-block; width: 15px; height: 10px; background-color: #00b3d9; margin-right: 5px;"></span> -10 to -15% cohort avg: 4 district(s)</div> <div><span style="display: inline-block; width: 15px; height: 10px; background-color: #a6d8e6; margin-right: 5px;"></span> -5 to -10% cohort avg: 12 district(s)</div> <div><span style="display: inline-block; width: 15px; height: 10px; background-color: white; border: 1px solid black; margin-right: 5px;"></span> Within <math>\pm 5\%</math> cohort avg: 16 district(s)</div> </div> <div style="width: 50%;"> <div><span style="display: inline-block; width: 15px; height: 10px; background-color: #f4a460; margin-right: 5px;"></span> +10 to +15% cohort avg: 5 district(s)</div> <div><span style="display: inline-block; width: 15px; height: 10px; background-color: #f9c996; margin-right: 5px;"></span> +15 to +20% cohort avg: 3 district(s)</div> <div><span style="display: inline-block; width: 15px; height: 10px; background-color: #fde08c; margin-right: 5px;"></span> <math>\geq +20\%</math> cohort avg: 10 district(s)</div> <div><span style="display: inline-block; width: 15px; height: 10px; background-color: #808080; margin-right: 5px;"></span> Missing data: 1 district(s)</div> </div> </div> | $\pm\%$ = Secondary regional district deviation (n=4) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|

Figure 5

## Appendix D. Maps from previous years (continued)

Geographic map showing 5-year PPE growth (2009-2014) vs enrollment cohort baseline

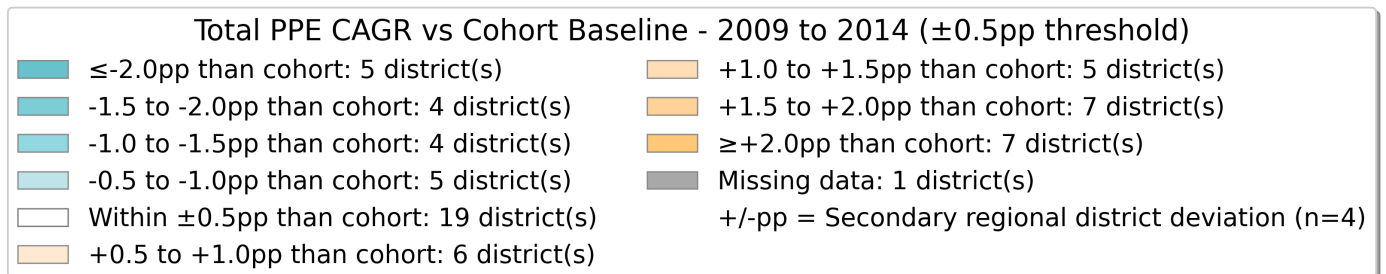
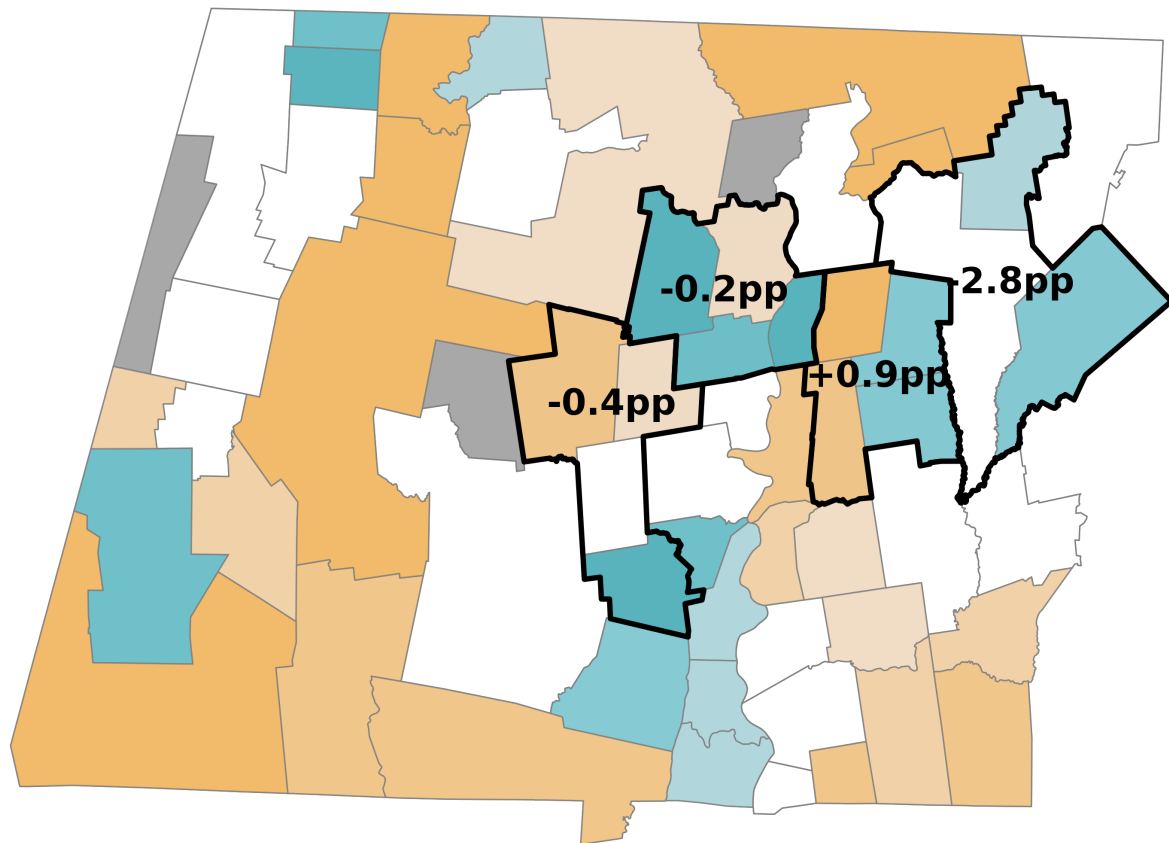


Figure 6

## Appendix D. Maps from previous years (continued)

Geographic map showing district locations and enrollment cohorts (2009)

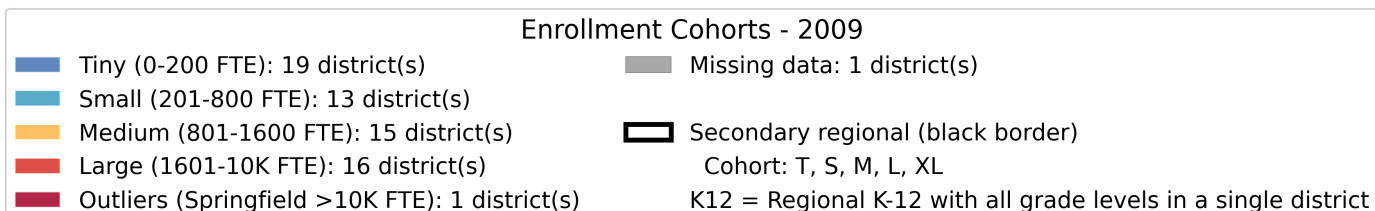
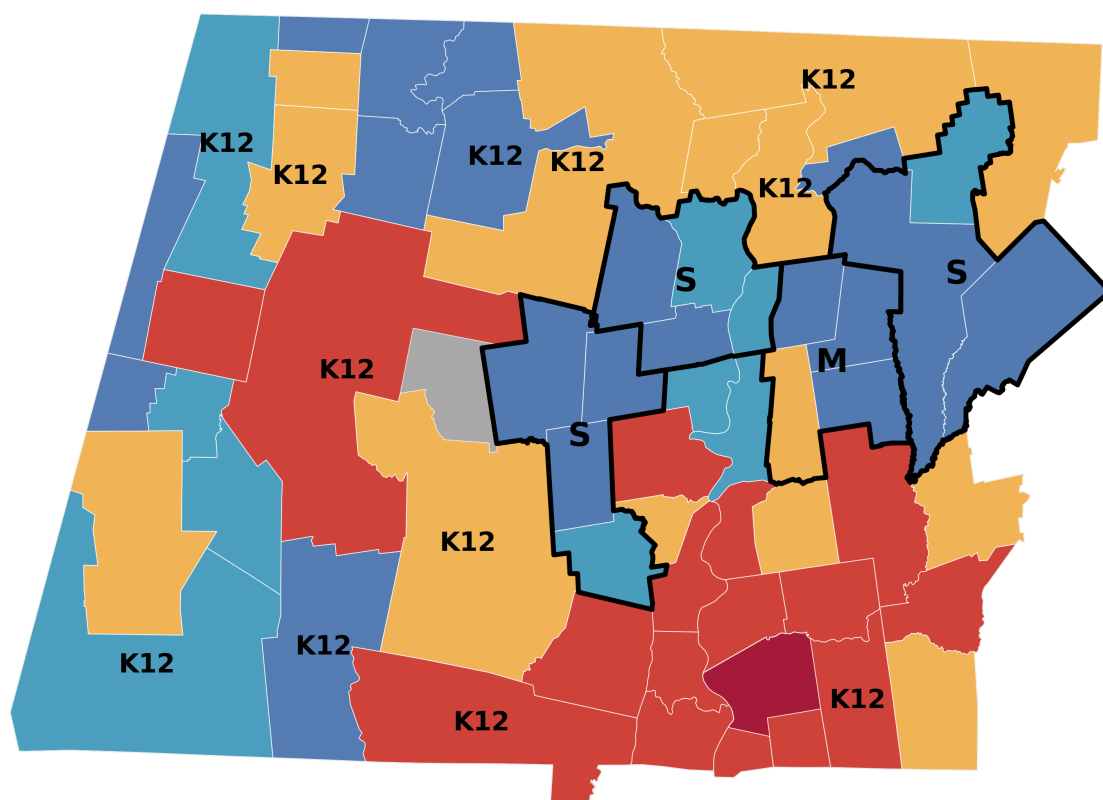
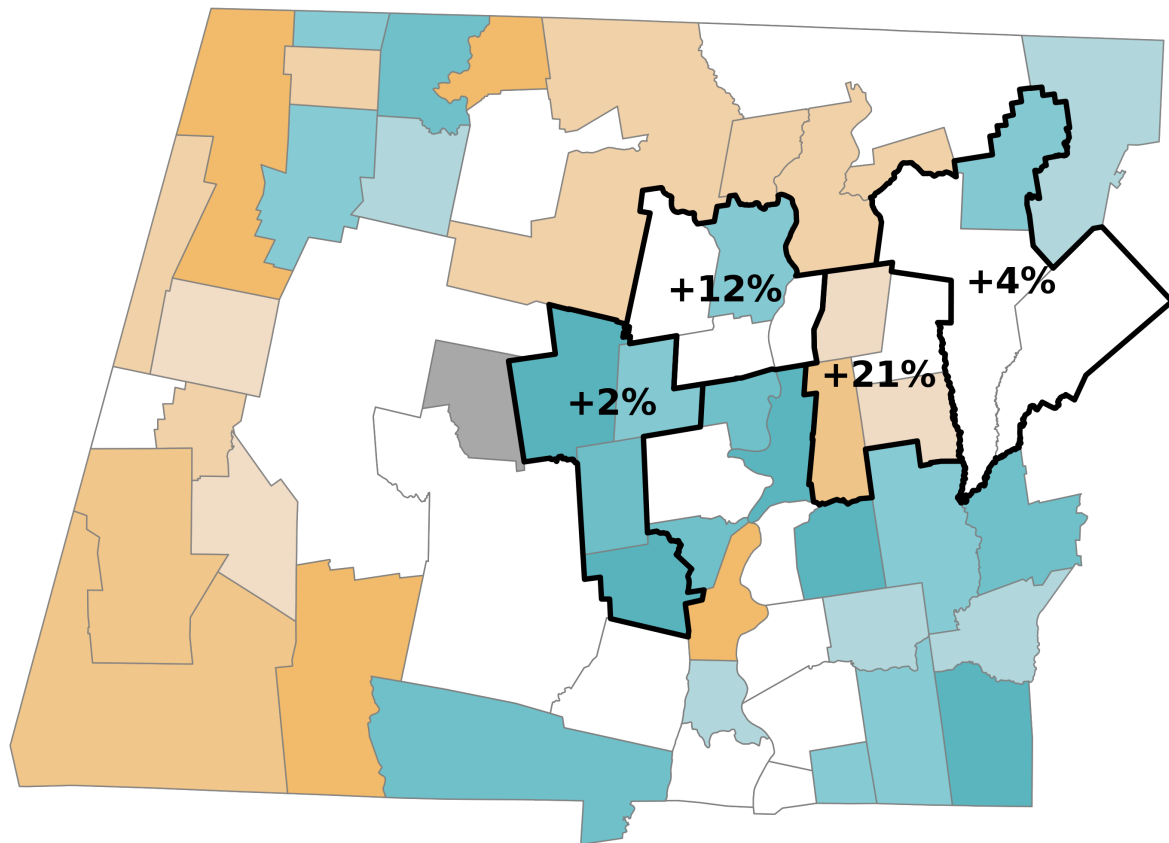


Figure 7

## Appendix D. Maps from previous years (continued)

Geographic map showing 2009 PPE vs enrollment cohort baseline



### Total PPE vs Cohort Baseline - 2009 ( $\pm 5\%$ threshold)

|                                               |                                                     |
|-----------------------------------------------|-----------------------------------------------------|
| ■ $\leq -20\%$ cohort avg: 5 district(s)      | ■ +10 to +15% cohort avg: 8 district(s)             |
| ■ -15 to -20% cohort avg: 6 district(s)       | ■ +15 to +20% cohort avg: 3 district(s)             |
| ■ -10 to -15% cohort avg: 8 district(s)       | ■ $\geq +20\%$ cohort avg: 5 district(s)            |
| ■ -5 to -10% cohort avg: 5 district(s)        | ■ Missing data: 1 district(s)                       |
| □ Within $\pm 5\%$ cohort avg: 20 district(s) | +/- % = Secondary regional district deviation (n=4) |
| ■ +5 to +10% cohort avg: 4 district(s)        |                                                     |

Figure 8